MATHEMATICS OF CLASSICAL AND QUANTUM PHYSICS

FREDERICK W. BYRON, JR.

AND

ROBERT W. FULLER

TWO VOLUMES BOUND AS ONE

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To Edith and Ann

PREFACE

This book is designed as a companion to the graduate level physics texts on classical mechanics, electricity, magnetism, and quantum mechanics. It grows out of a course given at Columbia University and taken by virtually all first year graduate students as a fourth basic course, thereby eliminating the need to cover this mathematical material in a piecemeal fashion within the physics courses. The two volumes into which the book is divided correspond roughly to the two semesters of the full-year course. The consolidation of the mathematics needed for graduate physics into a single course permits a unified treatment applicable to many branches of physics. At the same time the fragments of mathematical knowledge possesed by the student can be pulled together and organized in a way that is especially relevant to physics. The central unifying theme about which this book is organized is the concept of a *vector space*. To demonstrate the role of mathematics in physics, we have included numerous physical applications in the body of the text, as well as many problems of a physical nature.

Although the book is designed as a textbook to complement the basic physics courses, it aims at something more than just equipping the physicist with the mathematical techniques he needs in courses. The mathematics used in physics has changed greatly in the last forty years. It is certain to change even more rapidly during the working lifetime of physicists being educated today. Thus, the physicist must have an acquaintance with abstract mathematics if he is to keep up with his own field as the mathematical language in which it is expressed changes. It is one of the purposes of this book to introduce the physicist to the language and the style of mathematics as well as the content of those particular subjects which have contemporary relevance in physics.

The book is essentially self-contained, assuming only the standard undergraduate preparation in physics and mathematics; that is, intermediate mechanics, electricity and magnetism, introductory quantum mechanics, advanced calculus and differential equations. The level of mathematical rigor is generally comparable to that typical of mathematical texts, but not uniformly so. The degree of rigor and abstraction varies with the subject. The topics treated are of varied subtlety and mathematical sophistication, and a logical completeness that is illuminating in one topic would be tedious in another.

While it is certainly true that one does not need to be able to follow the proof of Weierstrass's theorem or the Cauchy–Goursat theorem in order to be able to

compute Fourier coefficients or perform residue integrals, we feel that the student who has studied these proofs will stand a better chance of growing mathematically *after* his formal coursework has ended. No reference work, let alone a text, can cover all the mathematical results that a student will need. What *is* perhaps possible, is to generate in the student the confidence that he can find what he needs in the mathematical literature, and that he can understand it and use it. It is our aim to treat the limited number of subjects we do treat in enough detail so that after reading this book physics students will not hesitate to make direct use of the mathematical literature in their research.

The backbone of the book—the theory of vector spaces—is in Chapters 3, 4, and 5. Our presentation of this material has been greatly influenced by P. R. Halmos's text, *Finite-Dimensional Vector Spaces.* A generation of theoretical physicists has learned its vector space theory from this book. Halmos's organization of the theory of vector spaces has become so second-nature that it is impossible to acknowledge adequately his influence.

Chapters 1 and 2 are devoted primarily to the mathematics of classical physics. Chapter 1 is designed both as a review of well-known things and as an introduction of things to come. Vectors are treated in their familiar three-dimensional setting, while notation and terminology are introduced, preparing the way for subsequent generalization to abstract vectors in a vector space. In Chapter 2 we detour slightly in order to cover the mathematics of classical mechanics and develop the variational concepts which we shall use later. Chapters 3 and 4 cover the theory of finite dimensional vector spaces and operators in a way that leads, without need for subsequent revision, to infinite dimensional vector spaces (Hilbert space)—the mathematical setting of quantum mechanics. Hilbert space, the subject of Chapter 5, also provides a very convenient and unifying framework for the discussion of many of the special functions of mathematical physics. Chapter 6 on analytic function theory marks an interlude in which we establish techniques and results that are required in all branches of mathematical physics. The theme of vector spaces is interrupted in this chapter, but the relevance to physics does not diminish. Then in Chapters 7, 8, and 9 we introduce the student to several of the most important techniques of theoretical physics—the Green's function method of solving differential and partial differential equations and the theory of integral equations. Finally, in Chapter 10 we give an introduction to a subject of ever increasing importance in physics—the theory of groups.

A special effort has been made to make the problems a useful adjunct to the text. We believe that only through a concerted attack on interesting problems can a student really "learn" any subject, so we have tried to provide a large selection of problems at the end of each chapter, some illustrating or extending mathematical points, others stressing physical applications of techniques developed in the text. In the later chapters of the book, some rather significant results are left as problems or even as a programmed series of problems, on the theory that as the student develops confidence and sophistication in the early chapters he will be able, with a few hints, to obtain some nontrivial results for himself.

The text may easily be adapted for a one-semester course at the graduate (or advanced undergraduate) level by omitting certain chapters of the instructor's choosing. For example, a one-semester course could be based on Volume 1. Another possibility, and one essentially used by one of the authors at the University of California at Berkeley, is to give a semester course based on the material in Chapters 3, 4, 5, and 10. On the other hand, a one-semester course in advanced mathematical methods in physics could be constructed from Volume II.

Certain sections within a chapter which are difficult and inessential to most of the rest of the book are marked with an asterisk.

In writing a book of this kind one's debts proliferate in all directions. In addition to the book of Halmos, we have been influenced by Courant–Hilbert's treatment of, and T. D. Lee's lecture notes on, Hilbert space, Riesz and Nagy's treatment of integral equations, and M. Hamermesh's book, *Group Theory*.

A special debt of gratitude is owed to R. Friedberg whose comments on the material have been extremely helpful. In particular, the presentation of Section 5.10 is based on his lecture notes.

Parts of the manuscript have also been read and taught by Ann L. Fuller, and her comments have improved it greatly. Richard Haglund and Steven Lundeen read and commented on the manuscript. Their painstaking work has removed many blemishes, and we thank them most sincerely.

Much of this book appeared in the form of lecture notes at Columbia University. Thanks are owed to the many students there, and elsewhere, who pointed out errors, or otherwise helped to improve the manuscript. Also, the enthusiasm of the students studying this material at Berkeley provided important encouragement.

While all the above named people have helped us to improve the manuscript, we alone are responsible for the errors and inadequacies that remain. We will be grateful if readers will bring errors to our attention so corrections can be made in subsequent printings.

One of us (FWB) held an Alfred P. Sloan Fellowship during much of the period of the writing; he gratefully thanks Professors M. Demeur and C. J. Joachain for their hospitality at the Université Libre de Bruxelles. The other author (RWF) would like to thank R. A. Rosenbaum of Wesleyan University, the University's Center for Advanced Studies, and its director, Paul Horgan, for their hospitality during the course of much of the work. We would also like to thank F. J. Milford and Battelle Memorial Institute's Seattle Research Center for providing support that facilitated the completion of the work.

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VOLUME ONE

CHAPTER 1

VECTORS IN CLASSICAL PHYSICS

INTRODUCTION

In this chapter we shall review informally the properties of the vectors and vector fields that occur in classical physics. But we shall do so in a way, and in a notation, that leads to the more abstract discussion of vectors in later chapters. The aim here is to bridge the gap between classical three-dimensional vector analysis and the formulation of abstract vector spaces, which is the mathematical language of quantum physics. Many of the ideas that will be developed more abstractly and thoroughly in later chapters will be anticipated in the familiar three-dimensional setting here. This should provide the subsequent treatment with more intuitive content. This chapter will also provide a brief recapitulation of classical physics, much of which can be elegantly stated in the language of vector analysis—which was, of course, devised expressly for this purpose. Our purpose here is one of informal introduction and review; accordingly, the mathematical development will not be as rigorous as in subsequent chapters.

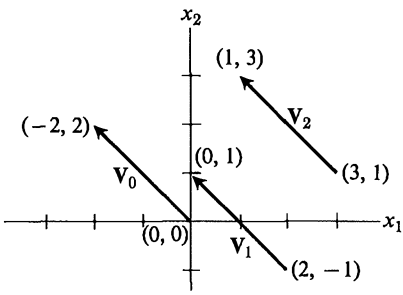

Fig. 1.1 Three equivalent vectors in a two-dimensional space.

1.1 GEOMETRIC AND ALGEBRAIC DEFINITIONS OF A VECTOR

In elementary physics courses the geometric aspect of vectors is emphasized. A vector, $\mathbf{x}$, is first conceived as a directed line segment, or a quantity with both a magnitude and a direction, such as a velocity or a force. A vector is thus distinguished from a scalar, a quantity which has only magnitude such as temperature, entropy, or mass. In the two-dimensional space depicted in Fig. 1.1, three vectors of equal magnitude and direction are shown. They form an

equivalence class which may be represented by $\mathbf{V}_0$, the unique vector whose initial point is at the origin. We shall gradually replace this elementary characterization of vectors and scalars with a more fundamental one. But first we must develop another language with which to discuss vectors.

An algebraic aspect of a vector is suggested by the one-to-one correspondence between the unique vectors (issuing from the origin) that represent equivalence classes of vectors, and the coordinates of their terminal points, the ordered pairs of real numbers (x_1, x_2). Similarly, in three-dimensional space we associate a geometrical vector with an ordered triple of real numbers, (x_1, x_2, x_3), which are called the *components* of the vector. We may write this vector more briefly as x_i, where it is understood that i extends from 1 to 3. In spaces of dimension greater than three we rely increasingly on the algebraic notion of a vector, as an ordered n-tuple of real numbers, $(x_1, x_2, \cdots, x_n)$. But even though we can no longer construct physical vectors for n greater than three, we retain the geometrical language for these n-dimensional generalizations. A formal treatment of the properties of such abstract vectors, which are important in the theory of relativity and quantum mechanics, will be the subject of Chapters 3 and 4. In this chapter we shall restrict our attention to the three-dimensional case.

There are then these two complementary aspects of a vector: the geometric, or physical, and the algebraic. These correspond to plane (or solid) geometry and analytic geometry. The geometric aspect was discovered first and stood alone for centuries until Descartes discovered algebraic or analytic geometry. Anything that can be proved geometrically can be proved algebraically and vice-versa, but the proof of a given proposition may be far easier in one language than in the other.

Thus the algebraic language is more than a simple alternative to the geometric language. It allows us to formulate certain questions more easily than we could in the geometric language. For example, the tangent to a curve at a point can be defined very simply in the algebraic language, thus facilitating further study of the whole range of problems surrounding this important concept. It is from just this formulation of the problem of tangents that the calculus arose.

It is said of Niels Bohr that he never felt he understood philosophical ideas until he had discussed them with himself in German, French, and English as well as in his native Danish. Similarly, one's understanding of geometry is strengthened when one can view the basic theorems from both the geometric and the algebraic points of view. The same is true of the study of vectors. It is all too easy to rely on the algebraic language to carry one through vector analysis, skipping blithely over the physical, geometric interpretation of the differential operators. We shall try to bring out the physical meanings of these operators as well as review their algebraic manipulation. The basic operators of vector analysis crop up everywhere in physics, so it pays to develop a physical picture of what these operators do—that is, what features they "measure" of the scalar or vector fields on which they operate.

1.2 THE RESOLUTION OF A VECTOR INTO COMPONENTS

One of the most important aspects of the study of vectors is the resolution of vectors into components. In fact, this will remain a central feature in Chapter 5, where we deal with Hilbert space, the infinite-dimensional generalization of a vector space. In three dimensions, any vector $\mathbf{x}$ can be expressed as a linear combination of any three noncoplanar vectors. Thus $\mathbf{x} = \alpha\mathbf{V}_1 + \beta\mathbf{V}_2 + \gamma\mathbf{V}_3$, where α, β, and γ are scalars. If we denote the length of a vector $\mathbf{x}$ by $|\mathbf{x}|$, then $\alpha|\mathbf{V}_1|$, $\beta|\mathbf{V}_2|$, and $\gamma|\mathbf{V}_3|$ are the components of $\mathbf{x}$ in the $\mathbf{V}_1$, $\mathbf{V}_2$, and $\mathbf{V}_3$ directions. The three vectors $\mathbf{V}_1$, $\mathbf{V}_2$, and $\mathbf{V}_3$ need not be perpendicular to each other—any three noncoplanar vectors form a base, or *basis*, in terms of which an arbitrary vector may be decomposed or expanded. But it is often most convenient to choose the basis vectors perpendicular to each other. In this case the basis is called *orthogonal*; otherwise it is called oblique. We shall deal almost exclusively with sets of orthogonal basis vectors.

A particularly useful set of basis vectors is the *Cartesian basis*, consisting of three mutually orthogonal vectors of unit length which have the same direction at all points in space. We shall denote unit vectors by the letter $\mathbf{e}$ in this chapter; accordingly, the Cartesian basis is the set $(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$ shown in Fig. 1.2. Such a set of base vectors is called *orthonormal*, because the vectors are orthogonal to each other and are normalized (have unit length). We shall not distinguish between a "basis" and a "coordinate system" in this treatment.

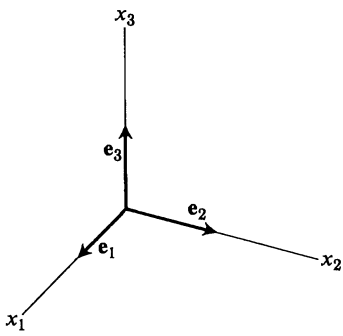

Fig. 1.2 Right-handed Cartesian basis or coordinate system.

The basis (or coordinate system) in Fig. 1.2 is *right-handed*, i.e., if the fingers of the right hand are extended along the positive x_1-axis and then curled toward the positive x_2-axis, the thumb will point in the positive x_3-direction. If any one of the basis vectors is reversed, we have a *left-handed* orthogonal basis. A mathematical definition of "handedness" will be given in Section 1.5.

An arbitrary vector may be expressed in terms of this Cartesian basis as $\mathbf{x} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + x_3\mathbf{e}_3$. The $\mathbf{e}_i$, or ith, component of $\mathbf{x}$ with respect to this basis is x_i, for $i = 1, 2, 3$. There are a great many other orthonormal bases, such as those of cylindrical and spherical and other curvilinear coordinate systems, which can greatly simplify the treatment of problems with special symmetry features. We shall deal with these in Section 1.7.

1.3 THE SCALAR PRODUCT

The scalar ("inner" or "dot") product of two vectors $\mathbf{x}$ and $\mathbf{y}$ is the real number defined in geometrical language by the equation

$$\mathbf{x}\cdot\mathbf{y} = |\mathbf{x}|\,|\mathbf{y}|\cos\theta\,,$$

where θ is the angle between the two vectors, measured from $\mathbf{x}$ to $\mathbf{y}$. Since $\cos\theta$ is an even function, the scalar product is commutative:

$$\mathbf{x}\cdot\mathbf{y} = \mathbf{y}\cdot\mathbf{x}\,.$$

Moreover, the scalar product is distributive with respect to addition:

$$\mathbf{x}\cdot(\mathbf{y}+\mathbf{z}) = \mathbf{x}\cdot\mathbf{y} + \mathbf{x}\cdot\mathbf{z}\,.$$

This equation has a familiar and reasonable appearance, but that is only because we automatically interpret it algebraically, where we usually take distributivity for granted. The reader will find it instructive to prove this by geometrical construction.

If $\mathbf{x}\cdot\mathbf{y} = 0$, it does not follow that one or both of the vectors are zero. It may be that they are perpendicular. Note that the length of a vector $\mathbf{x}$ is given by

$$|\mathbf{x}| \equiv x = (\mathbf{x}\cdot\mathbf{x})^{1/2}\,,$$

since $\cos\theta = 1$ for $\theta = 0$. In particular, for Cartesian basis vectors, we have

$$\mathbf{e}_i\cdot\mathbf{e}_j = \delta_{ij}\,, \tag{1.1}$$

where δ_{ij} is the Kronecker delta defined by

$$\delta_{ij} = \begin{Bmatrix} 1 & \text{if} & i=j \\ 0 & \text{if} & i\neq j \end{Bmatrix}. \tag{1.2}$$

If we expand two arbitrary vectors, $\mathbf{x}$ and $\mathbf{y}$, in terms of the Cartesian basis,

$$\mathbf{x} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + x_3\mathbf{e}_3 = \sum_{i=1}^{3} x_i\mathbf{e}_i\,,$$

$$\mathbf{y} = y_1\mathbf{e}_1 + y_2\mathbf{e}_2 + y_3\mathbf{e}_3 = \sum_{i=1}^{3} y_i\mathbf{e}_i\,,$$

then

$$\begin{aligned}\mathbf{x}\cdot\mathbf{y} &= \Big(\sum_i x_i\mathbf{e}_i\Big)\cdot\Big(\sum_j y_j\mathbf{e}_j\Big)\\ &= \sum_{i,j} x_i y_j\,\mathbf{e}_i\cdot\mathbf{e}_j = \sum_{i,j} x_i y_j\,\delta_{ij}\\ &= \sum_i x_i y_i\,.\end{aligned} \tag{1.3}$$

Here we have used the distributivity of the scalar product;

$$\sum_{i,j} \quad \text{stands for} \quad \sum_i\sum_j\,.$$

This last expression may be taken as the algebraic definition of the scalar product.

It follows that the length of a vector is given in terms of the scalar product by $|\mathbf{x}| = (\mathbf{x}\cdot\mathbf{x})^{1/2} = (\sum_i x_i^2)^{1/2}$. This equation provides an independent way of associating with any vector, a number called its length. We see that the notion of length need not be taken as inherent in the notion of vector, but is rather a consequence of defining a scalar product in a space of abstract vectors. Thus in Chapter 3 we shall study abstract vector spaces in which no notion of length has been defined. Then, in Chapter 4 we shall add an inner (or scalar) product to this vector space and focus on the enriched structure that results from this addition.

We shall now introduce a notational shorthand known as the *Einstein summation convention*. Einstein, in working with vectors and tensors, noticed that whenever there was a summation over a given subscript (or superscript), that subscript appeared *twice* in the summed expression, and vice versa. Thus one could simply omit the redundant summation signs, interpreting an expression like $x_i y_i$ to mean summation over the repeated subscript from 1 to, in our case, 3. If there are two distinct repeated subscripts, two summations are implied, and so on. In a letter, Einstein refers with tongue in cheek to this observation as "a great discovery in mathematics," but if you don't believe it is, just try getting along without it! (Another story in this connection—probably apocryphal—has it that the printer who was setting type for one of Einstein's papers noticed the redundancy and suggested omitting the summation signs.)

We shall adopt Einstein's summation convention throughout this chapter. In terms of this convention we have, for example,

$$\mathbf{x} = x_i\mathbf{e}_i\,,$$

$$\mathbf{x}\cdot\mathbf{y} = x_i y_j\,\delta_{ij} = x_i y_j = x_j y_j\,,$$

$$\mathbf{x}\cdot\mathbf{e}_i = x_j\mathbf{e}_j\cdot\mathbf{e}_i = x_j\,\delta_{ij} = x_i\,.$$

The last equation defines x_i, the component of $\mathbf{x}$ in the $\mathbf{e}_i$ direction, $\mathbf{x}\cdot\mathbf{e}_i$ is also called the *projection* of $\mathbf{x}$ on the $\mathbf{e}_i$ axis. The set of numbers $\{x_i\}$ is called the representation (or the *coordinates*) of the vector $\mathbf{x}$ in the basis (or the coordinate system) $\{\mathbf{e}_i\}$.

1.4 ROTATION OF THE COORDINATE SYSTEM: ORTHOGONAL TRANSFORMATIONS

We shall now consider the relationship between the components of a vector expressed with respect to two different Cartesian bases with the same origin, as shown in Fig. 1.3. Any vector $\mathbf{x}$ can be resolved into components with respect to either the K or the K' system. For example, in K we have

$$\mathbf{x} = (\mathbf{x}\cdot\mathbf{e}_j)\mathbf{e}_j = x_j\mathbf{e}_j\,, \tag{1.4}$$

where we are using the summation convention. In particular, if we take $\mathbf{x} = \mathbf{e}'_i$ $(i = 1, 2, 3)$, we can express the primed set of basis vectors in terms of the unprimed set:

$$\mathbf{e}'_i = (\mathbf{e}'_i \cdot \mathbf{e}_j)\mathbf{e}_j \equiv a_{ij}\mathbf{e}_j \qquad (j = 1, 2, 3) \ . \tag{1.5}$$

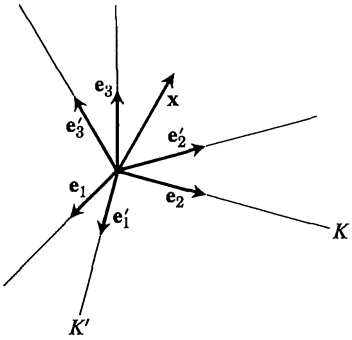

Fig. 1.3 Two different Cartesian bases with the same origin. The vector $\mathbf{x}$ can be expressed in terms of either basis.

The nine terms a_{ij} defined by Eq. (1.5), are the directional cosines of the angles between the six axes. These numbers may be written as the square array,

$$R \equiv (a_{ij}) \equiv \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}; \tag{1.6}$$

R is known as the *rotation matrix* in three dimensions, since it describes the consequences of a change from one basis to another (rotated) basis.

Note that in defining the matrix elements by the equation

$$a_{ij} \equiv \mathbf{e}'_i \cdot \mathbf{e}_j \ , \tag{1.7a}$$

we have adopted a certain convention. We could just as well have defined matrix elements a'_{ij} by

$$a'_{ij} \equiv \mathbf{e}'_j \cdot \mathbf{e}_i = a_{ji} \ . \tag{1.7b}$$

Almost all authors use the convention of Eq. (1.7a), but this is an arbitrary choice; a completely consistent development of the theory is possible based on the definition (1.7b). In fact, in abstract vector space theory, matrices are usually defined by a convention consistent with Eq. (1.7b) rather than by our rule, Eq. (1.7a). However, by replacing a_{ij} by a_{ji} in Eq. (1.6) and the equations we shall derive presently—thus interchanging rows and columns or *transposing* the matrix R—we may shuttle back and forth between conventions. In chapter 4 we shall reconsider these issues in a more general setting that permits an easy and complete systematization.

It is apparent that the elements of the rotation matrix are not independent. Since the basis vectors form an orthonormal set, it follows from Eq. (1.5) that

$$\delta_{ij} = \mathbf{e}'_i \cdot \mathbf{e}'_j = a_{ik}(\mathbf{e}_k \cdot \mathbf{e}'_j) = a_{ik}a_{jk} \ . \tag{1.8}$$

Equation (1.8) stands for a set of nine equations (of which only six are distinct), each involving a sum of three quadratic terms. It is left to the reader to show (by expanding the unprimed vectors in terms of the primed basis and taking scalar products) that we also have the relation

$$a_{ki}a_{kj} = \delta_{ij} . \tag{1.9}$$

The expressions (1.8) and (1.9) are referred to as orthogonality relations; the corresponding transformations (Eq. 1.5) are called *orthogonal transformations*.

In an n-dimensional space, the rotation matrix will have n^2 elements, upon which the orthogonality relations place $\frac{1}{2}(n^2 + n)$ conditions, as the reader can verify. Thus

$$n^2 - \tfrac{1}{2}(n^2 + n) = \tfrac{1}{2}n(n - 1)$$

of the a_{ij} are left undetermined. In a two-dimensional space this leaves one free parameter, which we may take as the angle of rotation. In a three-dimensional space there are three degrees of freedom, corresponding to the three so-called *Euler angles* used to describe the orientation of a rigid body.

Equation (1.5), together with the orthogonality relations, tells us how one set of orthogonal basis vectors is expressed in terms of another rotated set. Now we ask: How are the *components* of a vector in K related to the components of that vectors in K', and vice versa?

Any vector $\mathbf{x}$ may be expressed either in the K system as $\mathbf{x} = x_j\mathbf{e}_j$, or in the K' system as $\mathbf{x} = x'_i\mathbf{e}'_i$. Let us first express the x_j, the components of $\mathbf{x}$ with respect to the basis $\mathbf{e}_j$, in terms of the x'_i, the components of $\mathbf{x}$ with respect to the basis $\mathbf{e}'_i$. Using Eq. (1.5), we have

$$\mathbf{x} = x'_i\mathbf{e}'_i = x'_i a_{ij}\mathbf{e}_j = x_j\mathbf{e}_j . \tag{1.10}$$

Now, since the basis vectors are orthogonal, we may identify their coefficients in Eq. (1.10):

$$x_j = a_{ij}x'_i . \tag{1.11}$$

(More formally, if $\alpha_i\mathbf{e}_i = \beta_i\mathbf{e}_i$, then $(\alpha_i - \beta_i)\mathbf{e}_i = 0$. Since this is a sum, it does not follow automatically that $\alpha_i = \beta_i$. However, taking the scalar product with $\mathbf{e}_i$ gives $(\alpha_i - \beta_i)\delta_{ij} = 0$, whence $\alpha_j = \beta_j$.)

To derive the inverse transformation, one could, of course, repeat the above procedure, substituting for the unprimed vectors. That is, instead of using Eq. (1.5), one could use the corresponding relation for the unprimed basis vectors in terms of the primed:

$$\mathbf{e}_i = (\mathbf{e}_i \cdot \mathbf{e}'_j)\mathbf{e}'_j = a_{ji}\mathbf{e}'_j .$$

However, using the orthogonality relations, we can derive this result directly from Eq. (1.11). Multiplying it by a_{kj}, summing over j, and using Eq. (1.8), we have

$$a_{kj}x_j = a_{kj}a_{ij}x'_i = \delta_{ki}x'_i = x'_k , \tag{1.12}$$

which gives the primed components in terms of the unprimed components.

In summary, we have

$$\begin{gathered} \mathbf{x} = x_i'\mathbf{e}_i' = x_j\mathbf{e}_j\,, \\ \mathbf{e}_i' = a_{ij}\mathbf{e}_j\,, \qquad \mathbf{e}_i = a_{ji}\mathbf{e}_j' \\ x_i' = a_{ij}x_j\,, \qquad x_i = a_{ji}x_j' \\ a_{ik}a_{jk} = a_{ki}a_{kj} = \delta_{ij}\,. \end{gathered} \tag{1.13}$$

It should be understood that these equations refer to the components of *one* vector, $\mathbf{x}$, as expressed with respect to *two* different sets of basis vectors, $\mathbf{e}_i$ and $\mathbf{e}_i'$. Thus unprimed basis vectors can be expressed in terms of the other (primed) basis. Thus a_{ij} is the jth component of $\mathbf{e}_i'$, expressed with respect to the unprimed basis, and a_{ji} is the jth component of $\mathbf{e}_i$ expressed with respect to the primed basis.

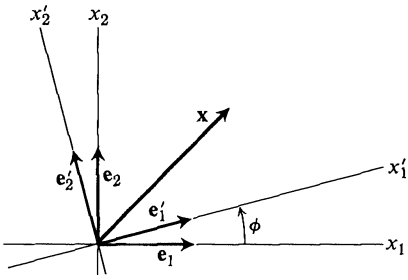

Fig. 1.4 Rotation in two dimensions, or rotation in three dimensions about an axis, x_3, orthogonal to the x_1, x_2, x_1', x_2' axes.

Example 1.1. *The two-dimensional rotation matrix.* We have defined the elements of the rotation matrix in Eqs. (1.5) and (1.6). For the two-dimensional case we have four coefficients: $a_{ij} \equiv (\mathbf{e}_i'\cdot\mathbf{e}_j)$, for $i, j = 1, 2$. From Fig. 1.4 it is clear that

$$(a_{ij}) = \begin{bmatrix} \cos\varphi & \sin\varphi \\ -\sin\varphi & \cos\varphi \end{bmatrix}. \tag{1.14a}$$

The first subscript of a_{ij} labels the row and the second subscript labels the column of the element a_{ij}. This rotation matrix tells us what happens to the components of a single vector $\mathbf{x}$ when we go from one basis, $\mathbf{e}_i$, to a new basis, $\mathbf{e}_i'$, by rotating the basis counterclockwise through an angle $(+\varphi)$. From Eq. (1.13), the components x_i of a vector $\mathbf{x}$ relative to the $\mathbf{e}_i$ basis and the components x_i' of that same vector relative to the $\mathbf{e}_i'$ basis are related by $x_i' = a_{ij}x_j$, or written out in full,

$$\begin{aligned} x_1' &= \cos\varphi x_1 + \sin\varphi x_2\,, \\ x_2' &= -\sin\varphi x_1 + \cos\varphi x_2\,. \end{aligned} \tag{1.15a}$$

Here x_i and x_i' refer to the components of a single vector with respect to two bases. The vector $\mathbf{x}$ sits *passively* as the basis with respect to which it is expressed rotates beneath it.

But there is another way to interpret these equations. We may regard the x_i as the components of one vector, $\mathbf{x}$, and the x_i' as the component of *another* vector, $\mathbf{x}'$, obtained from $\mathbf{x}$ by rotating $\mathbf{x}$ through the angle $(-\varphi)$. Let $X \equiv \{\mathbf{e}_i\}$ be the original set of basis vectors, and let $X' \equiv \{\mathbf{e}_i'\}$ denote the basis obtained by rotating X through the angle $(+\varphi)$. Then the components of $\mathbf{x}'$ referred to X are numerically equal to the components of $\mathbf{x}$ referred to X'. The idea behind this is actually quite simple; the difficulties are largely notational and can best be bypassed by drawing a diagram. The rotation of a vector through an angle $(-\varphi)$ produces a new vector with components in the original fixed basis equal to the components of the original vector, viewed as fixed, with respect to a new basis obtained from the original basis by rotating the original basis through the angle $(+\varphi)$.

Thus the equations that describe the *active* transformation of one vector into a new vector, rotated with respect to the original vector through an angle $(+\varphi)$, are obtained by substituting the angle $(-\varphi)$ into Eq. (1.15a), which gives

$$\begin{aligned} x_1' &= \cos\varphi x_1 - \sin\varphi x_2 \\ x_2' &= \sin\varphi x_1 + \cos\varphi x_2 \,. \end{aligned} \tag{1.16a}$$

Here the x_i and x_i' refer to the components of two vectors with respect to a single basis. Note that Eqs. (1.16a) may be written as

$$x_i' = a_{ji} x_j \,, \tag{1.16b}$$

where the a_{ij} are defined in Eq. (1.14a).

If we had defined the rotation matrix as the *transpose* of (a_{ij}), that is, if we had used Eq. (1.7b), then we would have

$$(a_{ij}') = \begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix} \tag{1.14b}$$

replacing Eq. (1.14a);

$$x_i' = a_{ji} x_j \qquad \text{or} \qquad x_i = a_{ij} x_j' \,, \tag{1.15b}$$

replacing Eq. (1.15a) for the components of a single vector in two different bases, and

$$x_i' = a_{ij} x_j \qquad \text{or} \qquad x_i = a_{ji} x_j' \,, \tag{1.16c}$$

replacing Eq. (1.16b), for the components of a transformed vector with respect to a single basis.

We may extend the rotation matrix (Eq. 1.14a) to the three-dimensional case of a rotation of basis vectors about the x_3-axis. Denoting this rotation matrix by $R(\varphi)$, we have

$$R(\varphi) \equiv (a_{ij}) = \begin{bmatrix} \cos\varphi & \sin\varphi & 0 \\ -\sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 1 \end{bmatrix} . \tag{1.17}$$

Example 1.2. *The three-dimensional rotation matrix* $R(\varphi, \theta, \psi)$. Suppose that we want to transform to a coordinate system in which the new z-axis, x_3', is in an arbitrarily specified direction, say along the vector **V** in Fig. 1.5. Such a rotation may be compounded of two three-dimensional rotations about an axis, such as those discussed in Example 1.1. First we rotate the coordinate system counterclockwise about the common x_3-x_3' axis through an angle φ. This gives

$$\mathbf{e}_i' = a_{ij}\mathbf{e}_j\,, \tag{1.18}$$

where the a_{ij} are given by Eq. (1.17). Now we rotate clockwise through an angle θ, that is, counterclockwise through the angle $(-\theta)$, in the $x_3'x_1'$-plane about the x_2'-axis. (We could as well have rotated about the x_1' axis but the sequence we have chosen is the conventional one.) The appropriate rotation matrix for this rotation of base about the x_2'-axis is

$$(b_{ij}) \equiv \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}, \tag{1.19}$$

and the new basis vectors, $\mathbf{e}_i''$, are given in terms of the primed ones by

$$\mathbf{e}_i'' = b_{ij}\mathbf{e}_j'\,.$$

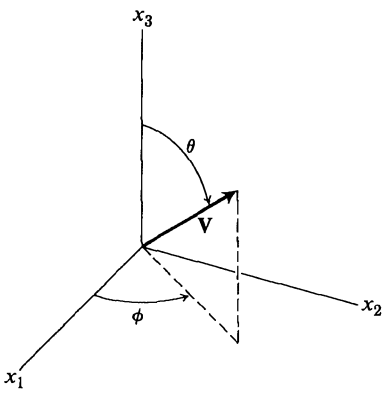

Fig. 1.5 The vector, **V**, which determines the z-axis of a rotated coordinate system.

Therefore, using Eq. (1.18), we have

$$\mathbf{e}_i'' = b_{ij}a_{jk}\mathbf{e}_k\,. \tag{1.20}$$

To go directly from the unprimed system to the doubly primed system we must know the coefficients c_{ik} in the equation

$$\mathbf{e}_i'' = c_{ik}\mathbf{e}_k\,. \tag{1.21}$$

Knowing these coefficients is equivalent to knowing the three-dimensional rotation matrix. From Eqs. (1.20) and (1.21), we see that

$$c_{ik} = b_{ij}a_{jk}\,. \tag{1.22}$$

Using this result and the matrices (1.17) and (1.19), we may compute the elements c_{ik}. The resulting rotation matrix is

$$R(\varphi, \theta) \equiv (c_{ik}) = \begin{bmatrix} \cos\varphi\cos\theta & \sin\varphi\cos\theta & -\sin\theta \\ -\sin\varphi & \cos\varphi & 0 \\ \cos\varphi\sin\theta & \sin\varphi\sin\theta & \cos\theta \end{bmatrix}. \tag{1.23}$$

This rotation matrix contains the matrix $R(\varphi)$ [Eq. (1.17)] as the special case $\theta = 0$; thus $R(\varphi, 0) = R(\varphi)$. Equation (1.22) is a special case as the general operation of matrix multiplication which will be treated fully in Chapter 3. The components of a vector $\mathbf{x}$ relative to the $\mathbf{e}_i$-basis and the components of that same vector $\mathbf{x}$ relative to the $\mathbf{e}_i''$-basis are related according to

$$x_i'' = c_{ij}x_j\,. \tag{1.24}$$

The rotation matrix $R(\varphi, \theta)$ does not represent the most general possible rotation. One more rotation is possible, a counterclockwise rotation through an angle ψ in the $x_1''x_2''$-plane about the x_3''-axis. This third rotation about an axis is described by the rotation matrix

$$(d_{ij}) = \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

And the grand "rotation of rotations," that may be achieved by compounding the three rotations about axes, through φ, θ, and ψ, is described by the rotation matrix whose elements are

$$[R(\varphi, \theta, \psi)]_{ij} = d_{ik}b_{kl}a_{lj}]\,.$$

The reader may verify that the resulting matrix is

$$R(\varphi, \theta, \psi) = \begin{bmatrix} \cos\varphi\cos\theta\cos\psi - \sin\varphi\sin\psi & \sin\varphi\cos\theta\cos\psi + \cos\varphi\sin\psi & -\sin\theta\cos\psi \\ -\cos\varphi\cos\theta\sin\psi - \sin\varphi\cos\psi & -\sin\varphi\cos\theta\sin\psi + \cos\varphi\cos\psi & \sin\theta\sin\psi \\ \cos\varphi\sin\theta & \sin\varphi\sin\theta & \cos\theta \end{bmatrix}. \tag{1.25}$$

The angles (φ, θ, ψ) are called the *Euler angles*. Their definition varies widely—the probability is small that two distinct authors' general rotation matrix will be the same. Note that $R(\varphi, \theta, 0) = R(\varphi, \theta)$ and $R(\varphi, 0, 0) = R(\varphi)$. The reader might note in passing that the determinant of $R(\varphi, \theta, \psi)$ (and all the other rotation matrices), has the value one. We shall prove this in Chapter 4.

Originally we introduced a vector as an ordered triple of numbers. The rule for expressing the components of a vector in one coordinate system in terms of its components in another system tells us that if we fix our attention on a physical vector and then rotate the coordinate system ($K \rightarrow K'$), the vector will have different numerical components in the rotated coordinate system. So we are led to realize that a vector is really more than an ordered triple. Rather,

it is many sets of ordered triples which are related in a definite way. One still specifies a vector by giving three ordered numbers, but these three numbers are distinguished from an arbitrary collection of three numbers by including the law of transformation under rotation of the coordinate frame as part of the definition. This law tells how all vectors change if the coordinate system changes. Thus one physical vector may be represented by infinitely many ordered triples. The particular triple depends on the orientation of the coordinate system of the observer. This is important because physical results must be the same regardless of one's vantage point, that is, regardless of the orientation of one's coordinate system. This will be the case if a given physical law involves vectors on both sides of the equation. Now, from this point of view, the transformation rule of Eq. (1.11) and the orthogonality relations, Eq. (1.9), may be used *to define* vectors. This is the natural starting point for a generalization to tensor analysis.

Since the orthogonal transformations are linear and homogeneous, it follows that the sum of two vectors is a vector and will transform acccording to Eq. (1.11) under orthogonal transformations. Also, if the equation $\mathbf{x} = \alpha\mathbf{y}$ (for example, $\mathbf{F} = m\mathbf{a}$), with α a scalar, holds in one coordinate system, it holds in any other which is related to the first by an orthogonal transformation. The reader may want to carry out the proofs of these statements formally.

We now prove a simple, but very important theorem.

Theorem. The scalar product is invariant under orthogonal transformations.

Proof. We see that this statement is obviously true when we consider the geometrical definition of the scalar product, for the lengths of vectors and the angle between them do not change as the axes are rotated. The algebraic proof is less transparent, but it allows some important generalizations. We have

$$\mathbf{x}' \cdot \mathbf{y}' = x_i' y_i' = a_{ij} x_j a_{ik} y_k = a_{ij} a_{ik} x_j y_k = \delta_{jk} x_j y_k = x_j y_j = \mathbf{x} \cdot \mathbf{y} ,$$

which completes the proof of the theorem.

Now scalars, ϕ, are invariant under rotations:

$$\phi' = \phi , \tag{1.26}$$

and the components of a vector transform according to

$$x_i' = a_{ij} x_j . \tag{1.27}$$

It is easy to generalize these notions and write down what is called a *Cartesian-tensor of the second rank*. In three dimensions, this is a set of nine components, T_{ij}, which under orthogonal transformations behave according to the rule

$$T_{ij}' = a_{im} a_{jn} T_{mn} .$$

A vector is a tensor of the first rank and a scalar is a tensor of zeroth rank. Generalization to tensors of higher rank is clearly possible, but we shall defer further discussion of tensors to Section 1.8.

The importance of thinking of these quantities in terms of their transformation properties lies in the requirement that physical theories must be in-

variant under rotation of the coordinate system. The inclination of the coordinate axes that we superimpose on a physical situation must not affect the physical answers we get. Or, to put it another way, observers who study a situation in different coordinate systems must agree on all physical results. For example, we may view the flight of a projectile (Fig. 1.6) from either the K or the K' system. In K, Newton's second law is $\mathbf{F} = m\mathbf{a}$, and the equations of motion are

$$m\ddot{x}_1 = 0, \qquad m\ddot{x}_2 = -mg\,.$$

Letting the initial ($t = 0$) conditions be

$$\begin{aligned} x_i(0) &= 0\,, \qquad i = 1, 2, \\ \dot{x}_1(0) &= v_0 \cos\theta\,, \\ \dot{x}_2(0) &= v_0 \sin\theta\,, \end{aligned}$$

we find for the trajectory in K,

$$x_2 = \frac{-g}{2v_0^2\cos^2\theta}\,x_1^2 + \tan\theta x_1\,.$$

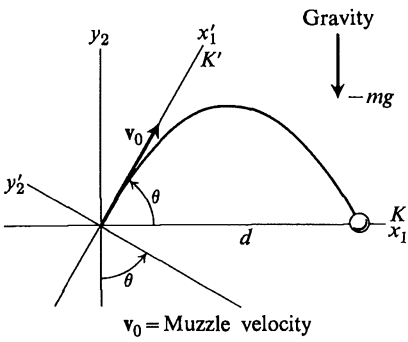

Fig. 1.6 The parabolic trajectory of a projectile viewed in coordinate systems K and K'.

In K', Newton's law is $F' = ma'$, and the equations of motion become

$$m\ddot{x}_1' = -mg\sin\theta\,, \qquad m\ddot{x}_2' = -mg\cos\theta\,.$$

The initial conditions become

$$\begin{aligned} x_i(0) &= 0\,, \qquad i = 1, 2, \\ \dot{x}_1(0) &= v_0\,, \\ \dot{x}_2(0) &= 0\,, \end{aligned}$$

and the trajectory is

$$x_1' = x_2' \tan\theta + v_0\left(\frac{2x_2'}{-g\cos\theta}\right)^{1/2}.$$

Do the trajectories as expressed in the primed and unprimed variables describe the same physical path? They had better! Otherwise Newton's law does not

provide a frame-independent description. It is left to the reader to reassure himself that all is well.

The important question is why this all works out. Just where is frame-independence for *rotated* coordinate systems built into Newton's laws? The answer is that two *vectors* which are equal in one frame, say K, are equal in a rotated frame, K'. The linear homogeneous character of the transformation law for vector components *guarantees* this. Instead of deriving the equations of motion in K' by looking at the physical situation in the frame, we could have derived them from the equations of motion as stated for K by applying the rotation matrix directly to the relevant vectors—forces and accelerations. It is instructive to carry this out once in one's life.

The key point is that on both sides of the equation there are *vector* quantities; hence under rotation of the basis vectors, both sides transform the same way. If on one side there were two numbers that remained constant under rotation (two such numbers would *not* be the components of a vector), while the other side was transforming like a vector, the equation would have a different form after transformation, and it would give different predictions. The world goes on independent of the inclination of our coordinate system, and we incorporate this isotropy of space into our theories from the start in the requirement that all terms in an equation be tensors of the same rank: all tensors of second rank, all vectors, or all scalars.

Another point worth noting is that since we get the same physical results in any frame, we can solve the problem in the frame where it is solved most easily—in our example, frame K. In general, we can establish a tensor equation in any particular frame and know immediately that it holds for every frame.

In summary then, the invariance of a physical law under orthogonal transformation of the spatial coordinate system requires that all the terms of the equation be tensors of the same rank. We say then that the terms are *covariant* under orthogonal transformations, i.e., they "vary together."

Later we shall view the Lorentz transformation of special relativity as an orthogonal transformation in four-dimensional space ("space-time" or Minkowski space), and again, we shall insist that all the terms of an equation be tensors (in this case, "four-tensors") of the same rank. This will ensure that the laws of physics are invariant under Lorentz transformations; that is, for all observers moving with uniform relative velocity.

1.5 THE VECTOR PRODUCT

The vector (or "cross") product of two vectors $\mathbf{x}$ and $\mathbf{y}$ is a vector—as we might expect—and is written $\mathbf{z} \equiv \mathbf{x} \times \mathbf{y}$. In geometrical language, we define the magnitude of the vector $\mathbf{z}$ by

$$|\mathbf{z}| = |\mathbf{x} \times \mathbf{y}| = |\mathbf{x}||\mathbf{y}| \sin \theta ,$$

where θ is the angle measured from $\mathbf{x}$ to $\mathbf{y}$ in such a way that $\theta \leq \pi$. $\mathbf{z}$ is defined to be perpendicular to the plane containing $\mathbf{x}$ and $\mathbf{y}$, and to point in a direction

given by the right-hand rule applied to $\mathbf{x}$ and $\mathbf{y}$, with fingers swinging in the direction of θ from $\mathbf{x}$ to $\mathbf{y}$, and the thumb giving the direction of $\mathbf{z}$. If one's thumb points "up" as one swings one's fingers from $\mathbf{x}$ to $\mathbf{y}$, then it will point "down" as one swings one's fingers from $\mathbf{y}$ to $\mathbf{x}$ (remember that $\theta \leq \pi$). Therefore, the vector product is anticommutative:

$$(\mathbf{x} \times \mathbf{y}) = -(\mathbf{y} \times \mathbf{x}) .$$

Therefore, $\mathbf{x} \times \mathbf{x} = \mathbf{0}$ (which is obvious geometrically, since $\sin 0 = 0$).

As an example of the vector product, consider the set of orthonormal basis vectors in the right-handed coordinate system of Fig. 1.2. It follows from the definition of the vector product that these basis vectors obey the relations

$$\mathbf{e}_i \times \mathbf{e}_j = \mathbf{e}_k , \tag{1.28}$$

where i, j, and k are any even permutation of the subscripts 1, 2, and 3. [Oddness and evenness of a permutation of (1, 2, 3) refer to the oddness or evenness of the number of interchanges of adjacent numbers needed to get to the order (1, 2, 3). Thus (3, 1, 2) is an even permutation and (3, 2, 1) is an odd permutation.] In this connection we define the symbol ε_{ijk} which will be useful in much the same way as the Kronecker delta:

$$\varepsilon_{ijk} = \begin{cases} +1 & \text{if } (i, j, k) \text{ is an even permutation of } (1, 2, 3) , \\ -1 & \text{if } (i, j, k) \text{ is an odd permutation of } (1, 2, 3) , \\ 0 & \text{otherwise (e.g., if 2 or more indices are equal)} . \end{cases} \tag{1.29}$$

There is, in fact, a very useful identity relating the ε_{ijk} symbol and the Kronecker delta. It is

$$\varepsilon_{ijk}\varepsilon_{klm} = \delta_{il}\delta_{jm} - \delta_{jl}\delta_{im} . \tag{1.30}$$

We leave the verification to the reader. It also follows immediately from Eq. (1.29) that

$$\varepsilon_{ijk} = \varepsilon_{kij} = \varepsilon_{jki} = -\varepsilon_{jik} = -\varepsilon_{kji} = -\varepsilon_{ikj} .$$

The handedness of a coordinate system may now be defined mathematically: a set of basis vectors $\mathbf{e}_i$ is said to form a right-handed Cartesian coordinate system if

$$\mathbf{e}_i \times \mathbf{e}_j = \varepsilon_{ijk}\mathbf{e}_k . \tag{1.31}$$

The coordinate system is left-handed if $\mathbf{e}_i \times \mathbf{e}_j = -\varepsilon_{ijk}\mathbf{e}_k$. Clearly, the replacement of any basis vector by its negative simply reverses the handedness of the coordinate system. We shall use right-handed coordinate systems throughout the book.

The algebraic definition of the vector product is

$$\mathbf{z} = \mathbf{x} \times \mathbf{y} = x_j\mathbf{e}_j \times y_k\mathbf{e}_k = x_jy_k\mathbf{e}_j \times \mathbf{e}_k = x_jy_k\varepsilon_{jki}\mathbf{e}_i = \varepsilon_{ijk}x_jy_k\mathbf{e}_i . \tag{1.32}$$

(Again, we have assumed that the vector product is distributive.) Thus the ith component of $\mathbf{z}$ is

$$z_i = \mathbf{z}\cdot\mathbf{e}_i = \varepsilon_{ijk}x_jy_k . \tag{1.33}$$

The proofs of the familiar vector identities become very simple in this notation.

Example 1.3. In order to establish the vector identity,

$$\mathbf{x} \times (\mathbf{y} \times \mathbf{z}) = \mathbf{y}(\mathbf{x}\cdot\mathbf{z}) - \mathbf{z}(\mathbf{x}\cdot\mathbf{y}) , \tag{1.34}$$

we must prove that the ith components of the two expressions are equal:

$$\begin{aligned}[\mathbf{x} \times (\mathbf{y} \times \mathbf{z})]_i &= \varepsilon_{ijk}x_j\varepsilon_{klm}y_l z_m \\ &= (\delta_{il}\delta_{jm} - \delta_{jl}\delta_{im})x_j y_l z_m \\ &= y_i x_j z_j - z_i x_j y_j \\ &= [\mathbf{y}(\mathbf{x}\cdot\mathbf{z}) - \mathbf{z}(\mathbf{x}\cdot\mathbf{y})]_i .\end{aligned}$$

Even easier is the proof of the identities

$$\mathbf{x}\cdot(\mathbf{y} \times \mathbf{z}) = \mathbf{z}\cdot(\mathbf{x} \times \mathbf{y}) = \mathbf{y}\cdot(\mathbf{z} \times \mathbf{x}) , \tag{1.35}$$

which we leave to the reader.

In Section 1.4, we derived the orthogonality relations

$$a_{ik}a_{jk} = a_{ki}a_{kj} = \delta_{ij} \tag{1.36}$$

by considering the scalar products of the basis vectors. As we noted, these relations among the elements of the rotation matrix mean that the a_{ij} are not independent of each other. A simple way to solve the system of Eqs. (1.36) is to take components of the vector product relations among the basis vectors. The lth component of

$$\mathbf{e}_i \times \mathbf{e}_j = \varepsilon_{ijk}\mathbf{e}_k$$

in the primed frame is

$$\varepsilon_{lmn}(\mathbf{e}_i\cdot\mathbf{e}'_m)\,(\mathbf{e}_j\cdot\mathbf{e}'_n) = \varepsilon_{ijk}(\mathbf{e}_k\cdot\mathbf{e}'_l) ;$$

here we are using Eq. (1.33) in the primed frame. Recalling that $a_{ij} = \mathbf{e}'_i\cdot\mathbf{e}_j$, we have immediately

$$\varepsilon_{ijk}a_{lk} = \varepsilon_{lmn}a_{mi}a_{nj} , \tag{1.37}$$

which gives any one of the a_{ij}'s in terms of the others, since upon choosing i and j there is only one term in the sum on the left.

We may use this result to prove that the vector product of two vectors does indeed transform like a vector. Suppose that two objects $\mathbf{x}$ and $\mathbf{y}$ transform as vectors under orthogonal transformations:

$$x'_j = a_{jn}x_n \qquad \text{and} \qquad y'_k = a_{km}y_m .$$

We must show that $\mathbf{z} = \mathbf{x} \times \mathbf{y}$ also transforms as a vector, i.e., that

$$z'_i \equiv (\mathbf{x}' \times \mathbf{y}')_i = a_{ij}z_j .$$

Using Eqs. (1.37), with suitably chosen dummy subscripts, one easily carries out the proof. We have

$$z'_i = \varepsilon_{ijk}x'_j y'_k = \varepsilon_{ijk}a_{jn}x_n a_{km}y_m = \varepsilon_{nml}a_{il}x_n y_m = a_{ij}\varepsilon_{jnm}x_n y_m = a_{ij}z_j .$$

Although the vector product transforms like a vector under rotations, it does not behave in all respects like an "ordinary" vector. After all, it is the *product* of two vectors. Each of the vectors that enter into the vector product changes sign when we replace all the basis vectors $\mathbf{e}_i$ by their negatives, $-\mathbf{e}_i$. This is called *inversion* of the coordinate system.

In two-dimensional space, inversion is just rotation by 180°. In three dimensions, however, inversion is equivalent to changing from a right-handed to a left-handed coordinate system. Analogous results hold for spaces of higher dimension: In even-dimensional spaces, inversion can be achieved by a rotation of basis vectors, whereas in odd-dimensional spaces, inversion requires the *reflection* of the basis vectors in the origin.

Vectors like the position $\mathbf{r}$ and the momentum $\mathbf{p}$ change sign under inversion. They are called *polar*, or ordinary, vectors. But a vector product of two polar vectors, such as $\mathbf{L} = \mathbf{r} \times \mathbf{p}$, will not change sign under inversion. Such vectors are called *axial* vectors or *pseudovectors*. The scalar product of a polar vector and a pseudovector is a *pseudoscalar*; it changes sign under an inversion, whereas a scalar does not.

Just as equations in classical physics cannot equate tensors of different rank because they have different transformation properties under *rotations*, we cannot equate pseudovectors to vectors or pseudoscalars to scalars, because they transform differently under *inversions*. For example, the equation

$$\frac{d\mathbf{L}}{dt} = \mathbf{r} \times \mathbf{F} = \mathbf{N}$$

relates the *two pseudovectors*, angular momentum and torque.

1.6 A VECTOR TREATMENT OF CLASSICAL ORBIT THEORY

In order to illustrate the power and use of vector methods, we shall employ them to work out the Keplerian orbits. This treatment is to be contrasted with that based on solving the differential equations of motion found, for example, in Chapter 3 of Goldstein.*

We first prove Kepler's second law which states that angular momentum is constant in a central force field. Differentiating the angular momentum

$$\mathbf{L} = \mathbf{r} \times \mathbf{p}$$

with respect to time, we obtain

$$\dot{\mathbf{L}} = \dot{\mathbf{r}} \times \mathbf{p} + \mathbf{r} \times \dot{\mathbf{p}}\,.$$

The first vector product vanishes because $\mathbf{p} = m\dot{\mathbf{r}}$ so that $\dot{\mathbf{r}}$ and $\mathbf{p}$ are parallel. The second vector product is simply $\mathbf{r} \times \mathbf{F}$ by Newton's second law, and hence vanishes for all forces directed along the position vector, that is, for all central forces. Thus $\mathbf{L} = \mathbf{r} \times m\mathbf{v}$ is a constant vector in central force motion. This implies that the position vector $\mathbf{r}$, and therefore the entire orbit, lies in a fixed

* H. Goldstein, *Classical Mechanics*. Reading, Mass.: Addison-Wesley, 1950.

plane in three-dimensional space. This result is essentially Kepler's second law, which is often stated in terms of the conservation of areal velocity, $|\mathbf{L}|/2m$.

We now turn to a familiar, but, in fact, very special central force, the inverse-square force of gravitation and electrostatics. For such a force, Newton's second law becomes

$$m\frac{d\mathbf{v}}{dt} = -\frac{k}{r^2}\mathbf{n}\,, \tag{1.38}$$

where $\mathbf{n} = \mathbf{r}/r$ is a unit vector in the $\mathbf{r}$-direction, and k is a constant; for the gravitational case it is Gm_1m_2, and for the electrostatic case, it is q_1q_2 in cgs units.

First we note that

$$\mathbf{v} = \dot{\mathbf{r}} = \dot{r}\mathbf{n} + r\dot{\mathbf{n}}\,,$$

and using this in the definition of angular momentum, we find that

$$\mathbf{L} = mr^2(\mathbf{n} \times \dot{\mathbf{n}})\,. \tag{1.39}$$

Now we consider

$$\begin{aligned}\frac{d}{dt}(\mathbf{v} \times \mathbf{L}) = \dot{\mathbf{v}} \times \mathbf{L} &= \frac{-k}{mr^2}(\mathbf{n} \times \mathbf{L}) \\ &= -k[\mathbf{n}\,(\dot{\mathbf{n}}\cdot\mathbf{n}) - \dot{\mathbf{n}}\,(\mathbf{n}\cdot\mathbf{n})]\,.\end{aligned}$$

Here we have used the constancy of $\mathbf{L}$, Eqs. (1.38) and (1.39), and the vector identity, Eq. (1.34). Since $\mathbf{n}\cdot\mathbf{n} = 1$, it follows by differentiation that $\dot{\mathbf{n}}\cdot\mathbf{n} = 0$. Thus we obtain

$$\frac{d}{dt}(\mathbf{v} \times \mathbf{L}) = k\dot{\mathbf{n}}\,,$$

and integrating, we have

$$\mathbf{v} \times \mathbf{L} = k\mathbf{n} + \mathbf{C}\,, \tag{1.40}$$

where $\mathbf{C}$ is a constant vector. Thus $\mathbf{C} = \mathbf{v} \times \mathbf{L} - k\mathbf{n}$ is a vector constant of the motion. It lies along, and fixes the position of, the major axis of the orbit, as we shall see after we complete the derivation of the orbit.

Now we form the scalar quantity

$$L^2 = \mathbf{L}\cdot(\mathbf{r} \times m\mathbf{v}) = m\mathbf{r}\cdot(\mathbf{v} \times \mathbf{L}) = mr\,(k + C\cos\theta)\,, \tag{1.41}$$

using Eqs. (1.35) and (1.40), where θ is the angle measured from $\mathbf{C}$ to $\mathbf{r}$. Solving for r, we obtain

$$r = \frac{L^2/km}{1 + C/k\cos\theta} = \frac{A}{1 + \varepsilon\cos\theta}\,. \tag{1.42}$$

This equation is the polar form for a conic section with one focus at the origin, where θ is measured from the constant vector $\mathbf{C}$ (which we may take to be the x-axis) to $\mathbf{r}$, and ε represents the *eccentricity* of the conic section; depending on

its values, the conic section is a hyperbola, parabola, ellipse, or circle. The eccentricity is easily determined in terms of the constants of motion:

$$\varepsilon = \frac{|\mathbf{C}|}{k} = \frac{1}{k}\,|(\mathbf{v} \times \mathbf{L}) - k\mathbf{n}|$$
$$= \frac{1}{k}[|\mathbf{v} \times \mathbf{L}|^2 + k^2 - 2k\mathbf{n}\cdot(\mathbf{v} \times \mathbf{L})]^{1/2}\,.$$

Now $|\mathbf{v} \times \mathbf{L}|^2 = v^2L^2$ because $\mathbf{v}$ is perpendicular to $\mathbf{L}$. Using Eq. (1.41), we obtain

$$\varepsilon = \frac{1}{k}\left[v^2L^2 + k^2 - \frac{2kL^2}{mr}\right]^{1/2} = \left[1 + \frac{2L^2}{mk^2}\left(\frac{m}{2}v^2 - \frac{k}{r}\right)\right]^{1/2} = \left[1 + \frac{2L^2E}{mk^2}\right]^{1/2},$$

where E is the constant energy of the system. We can now write out in full the orbit equation known as Kepler's first law:

$$r = \frac{L^2/km}{1 + (1 + 2L^2E/mk^2)^{1/2}\cos\theta}\,. \tag{1.43}$$

The vector $\mathbf{C}$ lies along the major axis in a direction from the focus to the perihelion, and its magnitude is

$$|\mathbf{C}| = k\varepsilon = k\,(1 + 2L^2E/mk^2)^{1/2}\,.$$

This is most easily seen by considering its value at the perihelion. Note that this vector vanishes for a circle ($\varepsilon = 0$) since there is no unique major axis. The existence of a further integral of the motion (besides energy and angular momentum) is due to degeneracy of the motion associated with potentials of the Coulomb form, k/r.* This degeneracy is perhaps more familiar in the analogous quantum-mechanical problem, the hydrogen atom, than in classical orbit theory. The above vector treatment is based on the existence of this special integral of the motion and is therefore not a general substitute for the usual methods based on the differential equations of motion.

1.7 DIFFERENTIAL OPERATIONS ON SCALAR AND VECTOR FIELDS

If to each point x_i ($i = 1, 2, 3$) in some region of space there corresponds a scalar, $\varphi(x_i)$, or a vector, $\mathbf{V}(x_i)$, we have a scalar or a vector field. Typical scalar fields are the temperature or density distribution in an object, or the electrostatic potential. Typical vector fields are the gravitational force, the velocity at each point in a moving fluid (e.g., a hurricane), or the magnetic-field intensity. Fields are functions defined at points of physical space, and may be time-dependent or time-independent.

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* There is an excellent discussion of this by David F. Greenberg in *Accidental Degeneracy*, *Am. J. Phys.* **34**, 1101 (1966). Our vector $\mathbf{C}$ is therein called by its historical name, the Runge-Lenz vector, and it is studied in a more general setting.

Table 1.1.

Name of coordinate system:	Cartesian	Cylindrical	Spherical	Parabolic*
q_1	x	r	r	ξ
q_2	y	θ	θ	η
q_3	z	z	φ	φ
h_1	1	1	1	$\sqrt{\xi^2+\eta^2}$
h_2	1	r	r	$\sqrt{\xi^2+\eta^2}$
h_3	1	1	$r\sin\theta$	$\xi\eta$

In this section we shall define various differential operations on scalar and vector fields—the gradient, divergence, curl, and Laplacian—and examine the form these operations take in general curvilinear coordinate systems. We shall also consider the aspect of the field that each of these operators "measures." If we know what information the different operations extract from given fields, we shall be able to understand more intuitively the physical content of the basic equations of classical physics.

For the solution of certain physical problems, spherical, cylindrical, or still other coordinates are often superior to Cartesian coordinates. For example, the problem of finding the electrostatic potential between two charged concentric spherical shells is much simpler in spherical coordinates than in Cartesian or cylindrical coordinates, because in spherical coordinates the potential can be expressed as a function of one of the three coordinates alone, namely, r.

It is possible to proceed formally from the definitions of the various differential operators in Cartesian coordinates and the coordinate transformation equations to expressions for these operators in the other coordinate systems. Alternatively, one can derive the required expressions from coordinate-free, geometrical definitions. We shall take the latter course, giving general derivations valid for any set of orthogonal curvilinear coordinates q_i related to the distance ds by the line element or "metric"

$$ds^2 = h_1^2\,dq_1^2 + h_2^2\,dq_2^2 + h_3^2\,dq_3^2\,. \tag{1.44}$$

If only one of the three orthogonal coordinates q_1, q_2, q_3 is varied, the corresponding line element may be written

$$ds_i = h_i\,dq_i\,, \qquad i = 1, 2, \text{ or } 3 \qquad \text{(no summation)}\,. \tag{1.45}$$

The three most common (and one less common) systems are shown in Table 1.1.

The Gradient

The gradient is a differential operator defined on a scalar field φ. The gradient of a scalar field, written grad φ, is a vector field defined by the requirement

* Parabolic coordinates are useful in the study of the quantum-mechanical hydrogen atom, and in investigations of the Stark effect.

that

$$\text{grad}\,\varphi \cdot d\mathbf{s} = d\varphi\,, \tag{1.46}$$

where $d\varphi$ is the differential change in φ corresponding to the arbitrary space displacement $d\mathbf{s}$. From this we see that

$$d\varphi = |\text{grad}\,\varphi|\;|d\mathbf{s}|\cos\theta\,, \tag{1.47}$$

where θ is the angle between the vector grad φ and the displacement vector $d\mathbf{s}$. Thus it is clear that the rate of change of φ is greatest if the differential displacement is in the direction of grad φ, when $\theta = 0$ and $\cos\theta$ has its maximum value, 1. This defines the direction of the vector grad φ at a point in space: It is the direction of maximum rate of change of φ from that point, i.e., the direction in which $d\varphi/ds$ is greatest. It also follows from Eq. (1.47) that the magnitude of the vector grad φ is simply this maximal rate of change, i.e. the rate of change $|d\varphi/ds|$ in the direction in which φ is changing most rapidly.

We may summarize this by saying that the gradient of φ is the directional derivative in the direction of the maximum rate of change of φ. For example, let φ be the elevation from sea level of points on the surface of a mountain, and therefore proportional to gravitational potential. The equipotentials are the lines of constant φ or constant altitude. Displacements along equipotentials produce no change in φ ($d\varphi = 0$), but displacements perpendicular to equipotentials are in the direction of most rapid change of altitude, and $d\varphi$ takes on its maximum value. The vector grad φ is always perpendicular to equipotential lines or surfaces.

The general form of the ith component of the gradient vector follows from the definition of the gradient, Eq. (1.46), and Eq. (1.45). We have

$$(\text{grad}\,\varphi)_i = \lim_{ds_i \to 0} \frac{\varphi(q_i + dq_i) - \varphi(q_i)}{ds_i} = \frac{1}{h_i}\frac{\partial\varphi}{\partial q_i}\,. \tag{1.48}$$

We shall use the abbreviation

$$\frac{\partial}{\partial q_i} \equiv \partial_i\,. \tag{1.49}$$

Therefore

$$(\text{grad}\,\varphi)_i = \frac{1}{h_i}\,\partial_i\varphi\,. \tag{1.50}$$

In Cartesian coordinates, where $h_i = 1$, this becomes

$$(\text{grad}\,\varphi)_i = \partial_i\varphi\,, \tag{1.51}$$

and consequently

$$\text{grad}\,\varphi = \mathbf{e}_i\partial_i\varphi\,, \tag{1.52}$$

where summation from 1 to 3 is implied. This equation is often written in terms of the symbolic vector operator, ∇, called the "del" operator; it is defined in

Cartesian coordinate systems by

$$\nabla_{\text{Cartesian}} \equiv \mathbf{e}_i \partial_i \,. \tag{1.53}$$

In terms of the del operator, we have, in Cartesian coordinate systems,

$$\text{grad}\, \varphi = \nabla_{\text{Cartesian}}\, \varphi \,. \tag{1.54}$$

The ∇ symbol is often used to denote the grad operator irrespective of the type of coordinates, in which case we have

$$\text{grad}\, \varphi \equiv \nabla\varphi = \frac{1}{h_1}(\partial_1\varphi)\mathbf{e}_1 + \frac{1}{h_2}(\partial_2\varphi)\mathbf{e}_2 + \frac{1}{h_3}(\partial_3\varphi)\mathbf{e}_3 \,, \tag{1.55}$$

where $\mathbf{e}_i$ is the unit vector corresponding to q_i in the positive q_i-direction; ∇ is an operator and should not be thought of as a vector. It differs from a vector just as d/dx differs from a number; ∇ acquires meaning only when operating on a scalar or vector function. Taken alone, it has no magnitude; however, it does transform properly under rotations, so it is sometimes treated formally as a vector.

The Divergence

The divergence is a differential operation defined on a vector field $\mathbf{V}$. The divergence of a vector field, written div $\mathbf{V}$, is a scalar field. It is defined at a point x_i in coordinate-free form as

$$\text{div}\, \mathbf{V} = \lim_{\Delta\tau \to 0} \frac{1}{\Delta\tau} \int_\sigma \mathbf{V} \cdot d\boldsymbol{\sigma} \,, \tag{1.56}$$

where $\Delta\tau$ is the volume enclosed by the surface σ, and $d\boldsymbol{\sigma} = \mathbf{n}\, d\sigma$, where $\mathbf{n}$ is the outward-directed unit vector normal to the "infinitesimal" element of surface $d\sigma$ surrounding the point x_i. We omit the demonstration that this definition is independent of the shape of $\Delta\tau$.

The physical meaning of the divergence of a vector field can be determined straight from this definition. The divergence at a point measures the flux of the vector field through an infinitesimal surface per unit volume, or the source strength per unit volume at a point.

We shall give some specific examples as soon as we informally prove Gauss's theorem. This result is almost immediately apparent from the definition, Eq. (1.56). We form the integral of div $\mathbf{V}$ over a finite volume τ. This region can be subdivided into arbitrarily many, arbitrarily small subregions, $\Delta\tau_i$. Each infinitesimal surface element $d\sigma$ belonging to a subregion $\Delta\tau_i$ in the interior of τ appears in two integrals (over two adjoining subregions). But since $\mathbf{n}$ is oppositely directed in these two internal surface elements, the integrals over them cancel. The only surface elements over which the integrals are not canceled by integrals over adjoining subregions lie on the surface σ of the finite volume τ. Thus we obtain Gauss's theorem:

$$\int_\tau \text{div}\, \mathbf{V}\, d\tau = \int_\sigma \mathbf{V} \cdot d\boldsymbol{\sigma} \,. \tag{1.57}$$

We now discuss an important example of the divergence operator which illustrates its physical meaning.

Example 1.4. Let the stationary volume τ, with surface σ, enclose a fluid of density ρ moving with velocity $\mathbf{v}$. The mass in τ will be

$$\int_\tau \rho \, d\tau ,$$

so the time rate of change of the mass in τ will be

$$\frac{\partial}{\partial t} \int_\tau \rho \, d\tau .$$

The mass of the fluid leaving τ per unit time can be calculated as

$$\int_\sigma \rho \mathbf{v} \cdot d\boldsymbol{\sigma} = \int_\tau \operatorname{div} (\rho \mathbf{v}) \, d\tau ,$$

where we have used the divergence theorem, Eq. (1.57). Now conservation of mass requires that

$$\frac{\partial}{\partial t} \int_\tau \rho \, d\tau + \int_\tau \operatorname{div} (\rho \mathbf{v}) \, d\tau = 0 ,$$

or

$$\int_\tau \left[\frac{\partial \rho}{\partial t} + \operatorname{div} (\rho \mathbf{v}) \right] d\tau = 0 .$$

For this to hold for an arbitrary volume τ, it must be true that

$$\frac{\partial \rho}{\partial t} + \operatorname{div} (\rho \mathbf{v}) = 0 . \tag{1.58}$$

This is the equation of continuity, or conservation of mass. It is just a restatement of conservation of mass, which was assumed in the derivation. A similar equation holds for charge in electromagnetism: ρ is then the charge density and $\rho \mathbf{v} = \mathbf{J}$ is the current density.

We now obtain an expression for the divergence of a vector field in orthogonal curvilinear coordinates from the integral definition, Eq. (1.56). We must calculate the integral of the outward normal component of $\mathbf{V}$, $\mathbf{V} \cdot \mathbf{n}$, over the surface of an infinitesimal volume $\Delta\tau = ds_1 \, ds_2 \, ds_3$. Let the volume be bounded by the six surfaces

$$q_i = a_i \qquad \text{and} \qquad q_i = a_i + \delta q_i , \qquad \text{for } i = 1, 2, \text{ and } 3 .$$

The integral of $\mathbf{V} \cdot \mathbf{n}$ over the surface of constant q_1 at the point $q_1 = a_1 + \delta q_1$ is given by

$$\int_{a_2}^{a_2+\delta q_2} dq_2 \int_{a_3}^{a_3+\delta q_3} dq_3 h_2(a_1 + \delta q_1, q_2, q_3) h_3(a_1 + \delta q_1, q_2, q_3) \, V_1(a_1 + \delta q_1, q_2, q_3)$$

(remember that, in this case, **n** points in the positive q_1-direction), while for the integral of $\mathbf{V}\cdot\mathbf{n}$ over the surface of constant q_1 at the point $q_1 = a_1$, we have

$$-\int_{a_2}^{a_2+\delta q_2} dq_2 \int_{a_3}^{a_3+\delta q_3} dq_3 h_2(a_1, q_2, q_3) h_3(a_1, q_2, q_3)\, V_1(a_1, q_2, q_3)$$

(here **n** points in the *negative* q_1 direction). Combining these two terms, we get

$$\int_{a_2}^{a_2+\delta q_2} dq_2 \int_{a_3}^{a_3+\delta q_3} dq_3 [V_1(a_1 + \delta q_1, q_2, q_3) h_2(a_1 + \delta q_1, q_2, q_3) h_3(a_1 + \delta q_1, q_2, q_3) - V_1(a_1, q_2, q_3) h_2(a_1, q_2, q_3) h_3(a_1, q_2, q_3)] .$$

Now, as δq_1 becomes small, this can be written as

$$\int_{a_2}^{a_2+\delta q_2} dq_2 \int_{a_3}^{a_3+\delta q_3} dq_3 \left[\frac{\partial}{\partial q_1}(V_1 h_2 h_3)\right]_{q_1=a_1} \delta q_1 .$$

Finally, as we let δq_2 and δq_3 become small, we can replace this expression by

$$\left[\frac{\partial}{\partial q_1}(V_1 h_2 h_3)\right]_{q_i=a_i} \delta q_1 \delta q_2 \delta q_3 .$$

Analogous expressions hold for the other two pairs of surfaces. Adding these three expressions and dividing both sides of Eq. (1.57) by

$$d\tau = h_1 h_2 h_3 \delta q_1 \delta q_2 \delta q_3 ,$$

we obtain a general formula for the divergence:

$$\operatorname{div} \mathbf{V} = \frac{1}{h_1 h_2 h_3}\left[\frac{\partial(h_2 h_3 V_1)}{\partial q_1} + \frac{\partial(h_3 h_1 V_2)}{\partial q_2} + \frac{\partial(h_1 h_2 V_3)}{\partial q_3}\right] . \tag{1.59}$$

As must be the case, the divergence of a vector field is a scalar field. In Cartesian coordinates, Eq. (1.59) reduces to

$$\operatorname{div} \mathbf{V} = \partial_i V_i = \nabla_{\text{Cartesian}} \cdot \mathbf{V} . \tag{1.60}$$

The scalar product of the ∇ symbol and a vector field is often used to denote div **V** regardless of the type of coordinates, in which case,

$$\operatorname{div} \mathbf{V} = \nabla\cdot\mathbf{V} = \frac{1}{h_1 h_2 h_3}\left[\frac{\partial(h_2 h_3 V_1)}{\partial q_1} + \frac{\partial(h_3 h_1 V_2)}{\partial q_2} + \frac{\partial(h_1 h_2 V_3)}{\partial q_3}\right] . \tag{1.61}$$

Note that the components of grad and div considered as vector operators are not equal in curvilinear coordinates, even though the symbol ∇ is often used in writing both operations.

The Curl

The curl is a differential operator defined on a vector field **V**, and is written curl **V**. The curl of a vector field is itself a vector field, and is defined in coordinate-free form as

$$(\operatorname{curl} \mathbf{V})\cdot\mathbf{n} = \lim_{\Delta\sigma\to 0} \frac{1}{\Delta\sigma} \oint_{\lambda} \mathbf{V}\cdot d\boldsymbol{\lambda} , \tag{1.62}$$

where $\Delta\sigma$ is the area of the surface bounded by the closed path λ, and $d\boldsymbol{\lambda} = \mathbf{t}\,d\lambda$, where $\mathbf{t}$ is a unit tangent vector along λ; $(\text{curl}\,\mathbf{V})\cdot\mathbf{n}$ is the component of curl $\mathbf{V}$ in the $\mathbf{n}$-direction, i.e., normal to the surface $\Delta\sigma$ in a direction given by the right-hand rule applied to the path of integration about λ. We omit the demonstration that surfaces of different shapes and orientations give equivalent results. The direction of curl $\mathbf{V}$ is given by the orientation of a plane surface, like the direction of the vector product. It will turn out that curl $\mathbf{V}$ is a pseudovector like the vector product.

Stokes's theorem follows immediately from the definition of the curl. We simply form the integral of curl $\mathbf{V}$ over a finite surface σ bounded by the finite curve λ. We subdivide the surface into arbitrarily many, arbitrarily small parts, each of which can be considered an infinitesimal plane surface. For each such surface we form the integrals in Eq. (1.62), and then add them. All contributions to line integrals arising from arcs of curves interior to the bounding perimeter λ are canceled by contributions from line integrals along arcs of adjoining infinitesimal plane surfaces, since $\mathbf{t}$ is oppositely directed along these arcs. The only arcs along which the integrals are not canceled by pairs lie on the bounding curve λ of the finite surface σ. Thus we obtain Stokes's theorem:

$$\int_\sigma (\text{curl}\,\mathbf{V})\cdot d\boldsymbol{\sigma} = \oint_\lambda \mathbf{V}\cdot d\boldsymbol{\lambda}\,, \tag{1.63}$$

where $d\boldsymbol{\sigma} = \mathbf{n}\,d\sigma$.

We shall now illustrate the physical meaning of the curl of a vector field with several examples.

Example 1.5. Consider a bucket of water rotating with constant angular velocity $\boldsymbol{\omega}$ about a central vertical axis $\mathbf{k}$ until the water surface stabilizes. We assume that the velocity of the water (in cylindrical coordinates) is $\mathbf{v} = \omega r\boldsymbol{\theta}$, where $\boldsymbol{\theta}$ is a unit vector in the θ-direction. Let us calculate curl $\mathbf{v}$ from Eq. (1.62), taking a circle of radius r for the curve λ.

$$\begin{aligned}\mathbf{k}\cdot\text{curl}\,\mathbf{v} &= \lim_{\Delta\sigma\to 0}\frac{1}{\Delta\sigma}\oint_\lambda \mathbf{v}\cdot d\boldsymbol{\lambda} = \lim_{r\to 0}\frac{1}{\pi r^2}\oint_\lambda \omega r\boldsymbol{\theta}\cdot\boldsymbol{\theta}\,d\lambda \\ &= \lim_{r\to 0}\frac{\omega r}{\pi r^2}\oint_\lambda d\lambda = \lim_{r\to 0}\frac{\omega r}{\pi r^2}(2\pi r) = 2\omega\,.\end{aligned}$$

The component of the curl in the $\mathbf{k}$-direction is twice the angular velocity. The curl of a velocity field is then a measure of the *angular* velocity of the fluid. If a miniature paddle wheel is placed anywhere in a velocity field, its angular velocity measures the curl of the field. If it does not rotate, the component of the curl in the direction perpendicular to the plane of the paddle wheel is zero. However, there may be curl despite straight-line flow. For example, water moving in a pipe will move more slowly along the wall of the pipe than water in the center, due to friction effects. Therefore a paddle wheel will be turned by the water even though the water always moves in straight lines. On the other hand, there need not necessarily be curl in curved-line flow. If a paddle wheel

is placed in an appropriately designed "whirlpool," in which the fluid moves with a velocity given by $\mathbf{v} = \boldsymbol{\theta} v_0(a/r)$ in cylindrical coordinates (where a, and v_0 are constants) the velocity field will have zero curl at all points $r \neq 0$, and the paddle wheel will not rotate. To show this we compute explicitly curl $\mathbf{v}$ at any point in the whirlpool different from $r = 0$; $\mathbf{v}$ is the velocity field of the fluid flow. The component of curl $\mathbf{v}$ in the $\mathbf{k}$-direction is

$$\mathbf{k}\cdot\text{curl}\,\mathbf{v} = \lim_{A\to 0}\frac{1}{A}\oint_{\lambda}\mathbf{v}\cdot d\boldsymbol{\lambda}\,,$$

where A is the area of the cork-shaped region (that contains the paddle wheel in Fig. 1.7) bounded by the four arcs: l_1, l_2, l_3, and l_4. The integral about the perimeter of A may be broken up into line integrals over these four arcs. The integrals over l_3 and l_4 vanish because $\mathbf{v}$ is perpendicular to $d\boldsymbol{\lambda}$ at every point on these arcs. Thus

$$\begin{aligned}\mathbf{k}\cdot\text{curl}\,\mathbf{v} &= \lim_{A\to 0}\frac{1}{A}\left[\int_{l_1}\mathbf{v}\cdot d\boldsymbol{\lambda} + \int_{l_2}\mathbf{v}\cdot d\boldsymbol{\lambda}\right]\\ &= \lim_{A\to 0}\frac{1}{A}\left[\int_{\theta_2}^{\theta_1}\frac{av_0}{r_1}\boldsymbol{\theta}\cdot\boldsymbol{\theta}r_1\,d\theta + \int_{\theta_1}^{\theta_2}\frac{av_0}{r_2}\boldsymbol{\theta}\cdot\boldsymbol{\theta}r_2\,d\theta\right] = 0\,,\end{aligned}$$

because the integrals are equal in magnitude and opposite in sign, the contributions along arcs l_1 and l_2 canceling. The x- and y-components of curl $\mathbf{v}$ are clearly zero.

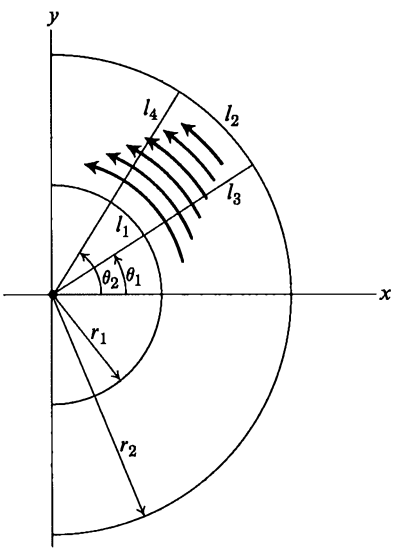

Fig. 1.7 The curl of a whirlpool. The flow lines are shown in boldface; the length of the lines indicates the speed of flow.

It is the purpose of streamlining to provide a surface past which air or water will flow with a minimum of curl, for motion with curl develops eddies that waste energy in heating up the air or water.

Of course a paddle wheel is appropriate only for fluid velocity fields. For electric fields, gravitational fields, etc., a "generalized paddle wheel," or curlmeter, must be visualized—one which responds to the forces associated with the particular vector field in question.

We complete our discussion of the curl operator, which up to now has been a coordinate-free treatment, by obtaining an expression for the components of the curl of a vector field in an orthogonal curvilinear coordinate system. To obtain the ith component of curl $\mathbf{V}$, we must calculate the line integral of the tangential component of the vector field $\mathbf{V}$, $\mathbf{V}\cdot\mathbf{t}$, along the boundary of an infinitesimal area $d\boldsymbol{\sigma}$ perpendicular to the ith basis vector. Consider an element of area in the q_2q_3-plane defined by $d\sigma = ds_2\,ds_3$ and oriented in the direction of increasing q_1. The 1-component of curl $\mathbf{V}$ is given by the sum of the line integrals, $\int \mathbf{V}\cdot d\boldsymbol{\lambda}$, along the four arcs that lead around $d\sigma$ from the point (q_2, q_3) to $(q_2 + dq_2, dq_3)$ to $(q_2 + dq_2, q_3 + dq_3)$ to $(q_2, q_3 + dq_3)$ back to (q_2, q_3). Thus, in brief, we have

$$(\text{curl } \mathbf{V})_1 = \lim_{d\sigma\to 0} \frac{1}{ds_2\,ds_3}[(V_2\,ds_2)_{q_3} + (V_3\,ds_3)_{q_2+dq_2} - (V_2\,ds_2)_{q_3+dq_3} - (V_3\,ds_3)_{q_2}]$$

$$= \lim_{ds_2\,ds_3\to 0}\left[\frac{(V_3\,ds_3)_{q_2+dq_2} - (V_3\,ds_3)_{q_2}}{ds_2\,ds_3} - \frac{(V_2\,ds_2)_{q_3+dq_3} - (V_2\,ds_2)_{q_3}}{ds_2\,ds_3}\right].$$

Therefore

$$(\text{curl } \mathbf{V})_1 = \frac{1}{h_2h_3}\left[\frac{\partial(h_3V_3)}{\partial q_2} - \frac{\partial(h_2V_2)}{\partial q_3}\right]. \tag{1.64}$$

The reader may want to amplify this argument in the fashion of the discussion that led to the expression for the divergence. The other two components of curl $\mathbf{V}$ may be obtained by cyclic permutation of the coordinate indices.

In Cartesian coordinates, Eq. (1.64) and the two additional equations for the other components reduce to

$$(\text{curl } \mathbf{V})_i = \varepsilon_{ijk}\partial_j V_k\,, \tag{1.65}$$

so in Cartesian coordinates,

$$\text{curl } \mathbf{V} = \varepsilon_{ijk}\mathbf{e}_i\partial_j V_k = \nabla_{\text{Cartesian}} \times \mathbf{V}\,. \tag{1.66}$$

The vector product of the ∇ symbol and a vector field is often used to denote curl $\mathbf{V}$, irrespective of the type of coordinates, so we may write

$$(\text{curl } \mathbf{V}) \equiv (\nabla \times \mathbf{V}) = \frac{1}{h_2h_3}[\partial_2(h_3V_3) - \partial_3(h_2V_2)]\mathbf{e}_1 + \frac{1}{h_3h_1}[\partial_3(h_1V_1)$$

$$- \partial_1(h_3V_3)]\mathbf{e}_2 + \frac{1}{h_1h_2}[\partial_1(h_2V_2) - \partial_2(h_1V_1)]\mathbf{e}_3\,, \tag{1.67}$$

where $\mathbf{e}_i$ is the unit vector corresponding to q_i.

The Laplacian

The Laplacian of a scalar field is defined as the divergence of the gradient of that scalar field. From Eqs. (1.50) and (1.61), we have the following expression for the Laplacian in orthogonal curvilinear coordinates:

$$\begin{aligned}\text{Laplacian } \varphi &\equiv \operatorname{div}(\operatorname{grad}\varphi) \equiv \nabla\cdot(\nabla\varphi) \equiv \nabla^2\varphi \\ &= \frac{1}{h_1h_2h_3}\left[\partial_1\left(\frac{h_2h_3}{h_1}\partial_1\varphi\right) + \partial_2\left(\frac{h_3h_1}{h_2}\partial_2\varphi\right) + \partial_3\left(\frac{h_1h_2}{h_3}\partial_3\varphi\right)\right]. \qquad (1.68)\end{aligned}$$

In Cartesian coordinates, this reduces to

$$\nabla^2_{\text{Cartesian}}\varphi = \partial_i\partial_i\varphi\,, \qquad (1.69)$$

where summation is implied.

Several of the most important partial differential equations of classical physics involve the Laplacian, e.g., Laplace's equation ($\nabla^2\varphi = 0$), Poisson's equation ($\nabla^2\varphi = \rho$), the heat and diffusion equations [$\nabla^2\varphi = K(\partial\varphi/\partial t)$], the wave equation [$\nabla^2\varphi + (1/c^2)(\partial^2\varphi/\partial t^2) = 0$], Helmholtz's equation ($\nabla^2\varphi + k^2\varphi = 0$), and others. Once we know what information the Laplacian extracts from a scalar field, we shall be able to intuit directly the physical meaning of these equations, which comprise a good part of classical physics.

To simplify the problem, we may use Cartesian coordinates without loss of generality. Let the scalar field φ have the value φ_0 at a certain point which we take as the origin of the Cartesian coordinate system. Consider a cube of side a, centred at the origin. The average value of φ in the cube is

$$\bar{\varphi} = \frac{1}{a^3}\int\int\int_{-a/2}^{a/2} \varphi\, dx_1\, dx_2\, dx_3\,, \qquad (1.70)$$

where we integrate over the interior of the cube. In order to evaluate the volume integral, we expand $\varphi(x_i)$ in a triple Taylor series in the three coordinates x_i, which we write as

$$\varphi(x_i) = \varphi_0 + (\partial_i\varphi)_0 x_i + \tfrac{1}{2}(\partial_i\partial_j\varphi)_0 x_i x_j + \cdots\,.$$

Substituting the Taylor expansion of $\varphi(x_i)$ into Eq. (1.70) and integrating from $-a/2$ to $a/2$, we see that the integral of any term that is linear in any x_i (in fact, any term containing an odd function of x_i) will vanish. The integrals of the quadratic (even) terms are easily carried out, and collecting them we obtain

$$\bar{\varphi} \cong \varphi_0 + \frac{a^2}{24}(\partial_i\partial_i\varphi)_0$$

or

$$(\nabla^2\varphi)_0 \cong \frac{24}{a^2}(\bar{\varphi} - \varphi_0)\,, \qquad (1.71)$$

with the approximation becoming better as the size of the cube (a) approaches zero. The value of the Laplacian of a scalar field at a point is therefore a

measure of the difference between the average value of the field in an infinitesimal neighborhood of that point and the value of the field at the point itself.

This result provides us with intuitive physical interpretations of the equations that contain the Laplacian operator. The simplest of these is Laplace's equation,

$$\nabla^2\varphi = 0\,, \tag{1.72}$$

which governs the scalar potential of the gravitational and electrostatic fields in the absence of mass or charge. We see that the average value of such a potential in a neighborhood of a point P must be equal to its value at P. A function with this property is called a *harmonic function*. We shall see in Chapter 6 that the real and imaginary parts of an analytic function are harmonic functions in two-dimensional space. Laplace's equation tells us immediately that a harmonic function cannot increase or decrease in all directions from a given point P, for then $\bar{\varphi}$ could not equal φ_P. If φ is a local maximum at P along a given line, it must be a local minimum along some other line through P; hence P must be a saddle point.

There also exist situations in which the value of a scalar at a point differs from the average value in the neighborhood of that point by an amount which is a function of space. This is the situation described by Poisson's equation:

$$\nabla^2\varphi = \rho(x_i)\,. \tag{1.73}$$

The scalar function $\rho(x_i)$ measures the density of mass or charge at the point x_i. Poisson's equation says that the departure of $\bar{\varphi}$ about x_i from $\varphi(x_i)$ is proportional to the density at x_i.

There are several physical processes in which the greater the difference between the valuc of φ_P and the average, $\bar{\varphi}$, in the neighborhood of P, the greater is the time rate of change of φ needed to equalize this difference. This is expressed mathematically by an equation known as the heat equation:

$$\nabla^2\varphi = K\frac{\partial\varphi}{\partial t}\,, \tag{1.74}$$

where K is a constant > 0. This equation governs heat-conduction processes (then φ is the temperature) and diffusion processes (then φ is the density of the diffusing material). The temperature or density changes with time at a rate proportional to the difference between its local and average values. The heat equation is a quantitative expression of Newton's law of cooling. Knowing what it is that the Laplacian operator does, we could go straight to an equation of the form of Eq. (1.74) if we were trying to express Newton's law of cooling mathematically.

Finally, the deviation of a local value from the average value can be proportional to the second time derivative, as in the wave equation:

$$\nabla^2\varphi = \frac{1}{c^2}\frac{\partial^2\varphi}{\partial t^2}\,. \tag{1.75}$$

The second derivative corresponds to accelerating the value of φ towards $\bar{\varphi}$, but in such a way that $\partial\varphi/\partial t \neq 0$ at equilibrium (i.e., at $\varphi_P = \bar{\varphi}$). The system moves through equilibrium and repeats its motion, like a simple harmonic oscillator.

We leave the interpretation of Helmholtz's equation

$$\nabla^2\varphi + k^2\varphi = 0 \tag{1.76}$$

to the reader (Problem 10).

Thus far we have regarded the Laplacian as operating only on a scalar field. it is useful to extend the application of this operator to vector fields in the following natural way. In Cartesian coordinates we define

$$\nabla^2_{\text{Cartesian}}\mathbf{V} \equiv \mathbf{e}_i\nabla^2 V_i \,. \tag{1.77}$$

Thus in Cartesian coordinates, the ith component of the vector $\nabla^2_{\text{Cartesian}}\mathbf{V}$ is the Laplacian of the ith component of $\mathbf{V}$, a scalar, defined in Eq. (1.69); that is

$$(\nabla^2_{\text{Cartesian}}\mathbf{V})_i = \partial_j\partial_j V_i \,. \tag{1.78}$$

To generalize the definition of the Laplacian of a vector to orthogonal curvilinear coordinates, we prove in Cartesian coordinates the vector identity

$$\nabla \times (\nabla \times \mathbf{V}) = -\nabla^2_{\text{Cartesian}}\mathbf{V} + \nabla(\nabla\cdot\mathbf{V}) \,, \tag{1.79}$$

which gives an expression for the curl of the curl of a vector field. Then we *define* the Laplacian of a vector in *curvilinear* coordinates, using Eq. (1.79), as

$$\nabla^2\mathbf{V} \equiv -\nabla \times (\nabla \times \mathbf{V}) + \nabla(\nabla\cdot\mathbf{V}) \,. \tag{1.80}$$

Since the curl, gradient, and divergence operators are all defined in curvilinear coordinates, and Eq. (1.80) holds in Cartesian coordinates, it is the natural definition of $\nabla^2\mathbf{V}$ in curvilinear coordinates. The proof of Eq. (1.79) parallels very closely that for the vector identity of Eq. (1.34), using the fact that in Cartesian coordinates,

$$[\nabla \times (\nabla \times \mathbf{V})]_i = \varepsilon_{ijk}\partial_j\varepsilon_{klm}\partial_l V_m \,.$$

In general, when dealing with the various differential operators, it is essential to keep firmly in mind the scalar or vector character of the field that results from the operation, and the scalar or vector character of the operand. The Laplacian is the only operator that is defined on both scalar and vector fields and it produces a scalar or vector field accordingly. The gradient operates on a scalar field and produces a vector field; the divergence operates on a vector field and produces a scalar field; and the curl operates on a vector field and produces a vector field.

There is a natural generalization of the Laplacian operator to four-dimensional space-time (Minkowski space) called the D'Alembertian operator, which is written $\Box^2$. It is defined in terms of the Laplacian as

$$\Box^2 \equiv \nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2} \,. \tag{1.81}$$

The significance of the minus sign will be discussed in Section 4.6. The operand of the D'Alembertian, as of the Laplacian, may be either a scalar or a vector field. In terms of the D'Alembertian, the scalar wave equation becomes simply

$$\Box^2\varphi = 0 . \tag{1.82}$$

The divergence of the vector grad φ is the Laplacian of φ. This suggests the following questions:

1. What is the curl of grad φ?
2. What is the divergence of the curl $\mathbf{V}$?

A direct calculation in Cartesian coordinates gives:

1. $$[\text{curl (grad } \varphi)]_i = \varepsilon_{ijk}\partial_j\partial_k\varphi = 0; \tag{1.83}$$

2. $$\text{div (curl } \mathbf{V}) = \partial_i\varepsilon_{ijk}\partial_j V_k = \varepsilon_{ijk}\partial_i\partial_j V_k = 0 . \tag{1.84}$$

These two proofs need a little explaining. In both we have the product of a term, ε_{ijk}, that is antisymmetric in any pair of subscripts, and a term of the form $\partial_m\partial_n$ which is symmetric in the subscripts. (A sufficient condition for the equality of cross derivatives, $\partial_m\partial_n\psi = \partial_n\partial_m\psi$, $m \neq n$, is their continuity, as is shown in any calculus text. We are assuming that the fields in physics possess this property.) Now the sum over the product of symmetric and antisymmetric terms vanishes. The reader can easily see this either by writing it out or by noting that if

$$a_{ij} = a_{ji} \qquad \text{and} \qquad b_{ij} = -b_{ji} ,$$

then

$$s \equiv a_{ij}b_{ij} = -a_{ji}b_{ji} = -s ,$$

since the subscripts are dummies; therefore $s = 0$.

These theorems hold in any curvilinear coordinate system. In fact, one way to show that div (curl $\mathbf{V}$) vanishes is to apply the coordinate-free statement of Gauss's theorem to a vector field derived from the curl of a vector field:

$$\int_\tau \text{div (curl } \mathbf{V})\, d\tau = \int_\sigma (\text{curl } \mathbf{V}) \cdot d\boldsymbol{\sigma} .$$

To evaluate the integral of curl $\mathbf{V}$ over the closed surface σ bounding τ, we draw a closed curve λ of any size and shape on the surface σ. This separates it into two surfaces, both bounded by λ. Now we use Stokes's theorem (Eq. 1.63) to evaluate the integral of curl $\mathbf{V}$ over these two surfaces. Since the two integrals have the common boundary λ which is traversed in a clockwise direction for one and a counterclockwise direction for the other, their values are equal in magnitude and opposite in sign. Hence their sum, which equals the surface integral of curl $\mathbf{V}$, is zero. Since this holds for any volume τ, it must be the case that div (curl $\mathbf{V}$) $= 0$.

The two theorems expressed in Eqs. (1.83) and (1.84) have very important converses. Namely, if curl $\mathbf{V} = \mathbf{0}$, then $\mathbf{V}$ may be expressed as the gradient of

a scalar field: $\mathbf{V} = \nabla\varphi$; and if div $\mathbf{V} = 0$, then $\mathbf{V}$ may be expressed as the curl of a vector field: $\mathbf{V} = \text{curl}\,\mathbf{A}$, where φ and $\mathbf{A}$ are referred to as the scalar and vector potential, respectively. Since our concern here is to gain an intuitive, physical picture of the differential operators and their role in classical physics, we shall not go into the proofs of these theorems, which are found in any book on vector analysis. A closely related result of potential theory is that one can determine a vector field uniquely from a knowledge of its curl and divergence, and suitable boundary conditions. Since Maxwell's equations are expressed in terms of the curl and divergence of the electromagnetic fields, it is essential to the program of classical electromagnetic field theory that this information be sufficient to solve for the fields themselves. This theorem is stated and proved in detail in Panofsky and Phillips.*

We conclude this section with a brief discussion of the self-consistency of Maxwell's equations in the light of the theorem that div (curl $\mathbf{V}$) $= 0$. This example also indicates what a compact and elegant notation the vector differential operators provide.

Most of the equations that bear Maxwell's name were already known when he came on the scene. What Maxwell did was to make a vital addition to one equation in the set of four empirical equations that described electrical and magnetic phenomena. This finishing touch filled in a missing symmetry in the equations and laid the basis for the prediction of wave phenomena. Prior to Maxwell, the following four laws were known. (We express them in differential form, in cgs units and standard notation.)

Coulomb's law: $$\nabla\cdot\mathbf{D} = 4\pi\rho \tag{1.85}$$

Nonexistence of magnetic monopoles: $$\nabla\cdot\mathbf{B} = 0 \tag{1.86}$$

Faraday's law (of electromagnetic induction): $$\nabla\times\mathbf{E} = -\frac{1}{c}\frac{\partial\mathbf{B}}{\partial t} \tag{1.87}$$

Ampere's law (for steady-state circuits): $$\nabla\times\mathbf{H} = \frac{4\pi}{c}\mathbf{J} \tag{1.88}$$

In Maxwell's day these equation were written out as cumbersome partial differential equations, which, of course, made it much harder to analyze them. In their present form any physics major could note, as Maxwell did, that Eq. (1.88) necessarily implies that div $\mathbf{J} = 0$. This equation holds only in the static case, with closed circuits; but if the charge is changing with time, and we assume charge conservation, then the continuity equation (1.58) holds. The continuity equation and Eq. (1.85) imply that

$$\nabla\cdot\mathbf{J} = -\frac{\partial\rho}{\partial t} = -\frac{\partial}{\partial t}\left(\frac{1}{4\pi}\nabla\cdot\mathbf{D}\right) = -\nabla\cdot\frac{1}{4\pi}\frac{\partial\mathbf{D}}{\partial t}\,.$$

* W. H. K. Panofsky and M. Phillips, *Classical Electricity and Magnetism.* Reading, Mass.: Addison-Welsey, 1955.

Thus if the term $(1/4\pi)(\partial \mathbf{D}/\partial t)$ is added to $\mathbf{J}$ on the right-hand side of Eq. (1.88), we would have

$$\nabla \times \mathbf{H} = \frac{4\pi}{c}\mathbf{J} + \frac{1}{c}\frac{\partial \mathbf{D}}{\partial t}, \tag{1.89}$$

this choice would ensure that $\nabla \cdot (\nabla \times \mathbf{H}) \equiv 0$; then the equations need no longer be restricted to the static case. The addition of this term, the so-called displacement current, was Maxwell's contribution. It fills in a missing symmetry in the equations, for they now express the fact that a time-dependent electric field generates a magnetic field, just as Faraday's law stated that a magnetic field changing in time generates an electric field. Therefore, the possibility of self-sustaining electromagnetic waves is suggested once Maxwell's new term is added to Eq. (1.88). Without the displacement-current term, the equations do not predict wave phenomena. The term is therefore of such tremendous significance that the whole set of equations is named for Maxwell.

1.8 CARTESIAN-TENSORS

Many physically significant quantities are neither scalars nor vectors. That is, the transformation law obeyed by these quantities under orthogonal transformations is neither

$$\phi' = \phi, \tag{1.90}$$

which defines a scalar [Eq. (1.26)], nor

$$x'_i = a_{ij}x_j, \tag{1.91}$$

which defines a vector [Eq. (1.27)].

If under an orthogonal transformation of coordinates in three dimensions, the 3^N quantities $T_{i_1 i_2 \cdots i_N}$ (where each of the N indices extends from 1 to 3) transform according to the rule

$$T'_{i_1 i_2 \cdots i_N} = a_{i_1 j_1} a_{i_2 j_2} \cdots a_{i_N j_N} T_{j_1 j_2 \cdots j_N}, \tag{1.92}$$

then $T_{i_1 i_2 \cdots i_N}$ are the components of an Nth-rank *tensor* (in three dimensions). Note that in Eq. (1.92), N summations are implied. Clearly a tensor of the first rank has three components that satisfy Eq. (1.91), so it is a vector. And a tensor of the zeroth rank has one invariant component and so it is a scalar. The most useful other case is the tensor of second rank. It has nine components, T_{ij}, which obey the transformation law

$$T'_{ij} = a_{ik} a_{jl} T_{kl}. \tag{1.93}$$

We shall discuss several applications of second-rank tensors in physics later in this section. But first it must be noted that what we are doing here really does not go under the name of tensor analysis. We are concerned here only with *Cartesian-tensors*, which are defined in terms of their transformation properties under a very special class of coordinate transformations, namely, ortho-

gonal transformations. Cartesian-tensors are sets of quantities with the transformation law, Eq. (1.92), under *orthogonal* transformations. As a general rule, *tensors* are sets of quantities with a certain law of transformation under *arbitrary* coordinate transformations, which is a much more stringent condition. Thus it is "easier" to qualify as a Cartesian-tensor than a tensor. Any tensor is a Cartesian-tensor, but a Cartesian-tensor need not be a tensor, because although a Cartesian-tensor obeys the proper transformation rule for the special orthogonal coordinate transformations, it may fail to do so for other coordinate transformations. This statement sounds paradoxical—grammatically it is, since on the surface it is a self-contradictory statement of the form: "Any tree is a maple tree, but a maple tree need not be a tree." We have tried to lessen this grammatical annoyance by writing "Cartesian-tensor" instead of "Cartesian tensor." The dash means that Cartesian-tensors are not a subclass of tensors, but new objects; the adjective does not select a subclass out of the class of objects called tensors.

We now discuss some notions of tensor analysis restricting our attention to three-dimensional Cartesian-tensors. However, we shall drop the qualifier "Cartesian," assuming that it is understood that all the tensors we refer to henceforth are, in fact, Cartesian-tensors.

First we consider the *contraction* of tensors, which is one of the most important operations defined on tensors. Contraction is an operation that produces a new tensor from an object whose tensor character is known. (In German the word for contraction is *Verjüngung*, which is translated as "rejuvenation.")

We shall need the result (whose proof we leave to the reader) that if $S_{i_1 i_2 \cdots i_N}$ is an Nth rank tensor and $T_{j_1 j_2 \cdots j_M}$ is an Mth rank tensor, then $S_{i_1 \cdots j_N} T_{j_1 \cdots j_M}$ has 3^{N+M} components and is an $(N+M)$th rank tensor.

Contraction of a tensor $T_{i_1 i_2 \cdots i_N}$ is defined as multiplication by, and thereby automatic summation over the indices of, the Kronecker delta, $\delta_{i_l i_m}$, where $l \neq m$, and l and $m \leq N$. Thus if $T_{i_1 \cdots i_N}$ is a tensor of Nth rank, then

$$H_{i_1 \cdots i_{j-1} i_{j+1} \cdots i_{m-1} i_{m+1} \cdots i_N} = \delta_{i_j i_m} T_{i_1 \cdots i_j \cdots i_m \cdots i_N} \tag{1.94}$$

is the contraction of T. We now prove that H is a tensor of rank $N-2$. First note that it has 3^{N-2} components, so that if it is a tensor, it is of rank $N-2$. Now

$$\begin{aligned} H'_{i_1 \cdots i_{j-1} i_{j+1} \cdots i_{m-1} i_{m+1} \cdots i_N} &= \delta'_{i_j i_m} T'_{i_1 \cdots i_j \cdots i_m \cdots i_N} \\ &= \delta_{i_j i_m} a_{i_1 n_1} \cdots a_{i_j n_j} \cdots a_{i_m n_m} \cdots a_{i_N n_N} T_{n_1 \cdots n_j \cdots n_m \cdots n_N} \\ &= a_{i_1 n_1} \cdots a_{i_m n_j} \cdots a_{i_m n_m} \cdots a_{i_N n_N} T_{n_1 \cdots n_j \cdots n_m \cdots n_N} . \end{aligned}$$

Since $a_{i_m n_j} a_{i_m n_m} = \delta_{n_j n_m}$, the last expression becomes, by Eq. (1.94),

$$\begin{aligned} &a_{i_1 n_1} \cdots a_{i_{j-1} n_{j-1}} a_{i_{j+1} n_{j+1}} \cdots a_{i_{m-1} n_{m-1}} a_{i_{m+1} n_{m+1}} \cdots \\ &\qquad a_{i_N n_N} H_{n_1 \cdots n_{j-1} n_{j+1} \cdots n_{m-1} n_{m+1} \cdots n_N} , \end{aligned}$$

which proves that H is a tensor (of rank $N-2$).

These general proofs for Nth rank tensors demand a certain virtuosity in subscript manipulation. Having indicated the general method in this one in-

stance, we shall now retreat to what is fortunately the most interesting physical case, $N = 2$.

By way of illustration, consider the product of two vectors, x_i and y_j. We know that the product $x_i y_j$ is a second-rank tensor, and we can contract it to get a quantity which, according to the theorem, must be a tensor of rank $2 - 2 = 0$ (i.e., a scalar). Thus

$$\delta_{ij} x_i y_j = x_i y_i = \mathbf{x} \cdot \mathbf{y} \,, \tag{1.95}$$

the scalar product.

Two important special types of second-rank tensors are the symmetric and antisymmetric tensors. Unlike vectors, tensors are not, in general, easily visualized geometrically, but for symmetric second-rank tensors ($T_{ij} = T_{ji}$) and antisymmetric second-rank tensors ($T_{ij} = -T_{ji}$) a geometrical representation is possible.

For an antisymmetric tensor of second rank, the three diagonal elements must be zero since $T_{ii} = -T_{ii}$. The other six elements occur in pairs of equal magnitude and opposite sign. Thus a tensor of this type is characterized by only three independent quantities (say T_{12}, T_{13}, and T_{23}), and a vector A_k can be associated with it according to the equation

$$A_i = \tfrac{1}{2} \varepsilon_{ijk} T_{jk}. \tag{1.96}$$

By comparing Eq. (1.96) with Eq. (1.33), we see that the vector associated with the second-rank antisymmetric tensor is in fact a pseudovector.

For a symmetric tensor of second rank, there are six independent components: the three diagonal elements, and the three pairs of equal off-diagonal elements. Identifying a second-rank symmetric tensor with a vector is therefore out of the question. However, it takes six independent parameters to specify a quadratic surface:

$$Ax_1^2 + Bx_2^2 + Cx_3^2 + Dx_1x_2 + Ex_1x_3 + Fx_2x_3 = 1 \,.$$

It is not surprising, therefore, that any symmetric second-rank tensor can be uniquely represented by the surface

$$T_{ij} x_i x_j = \pm 1 \,,$$

where the sign is that of the determinant of T_{ij}. The sign convention is needed, for otherwise, for the symmetric tensor $T_{ij} = -\delta_{ij}$ we would get the surface

$$-x_1^2 - x_2^2 - x_3^2 = 1 \,,$$

which does not exist in real space. With the sign convention, we get a sphere of radius one about the origin.

An arbitrary second-rank tensor, R_{ij}, can be written as the sum of a symmetric and an antisymmetric tensor:

$$R_{ij} = \underbrace{\tfrac{1}{2}(R_{ij} + R_{ji})}_{\text{symmetric}} + \underbrace{\tfrac{1}{2}(R_{ij} - R_{ji})}_{\text{antisymmetric}} \,. \tag{1.97}$$

It is not uncommon that after reading for the first time about tensors the student will say, "Yes, fine, but what *is* a tensor?" Providing him with a list of examples of tensors, and reiterating the definition in terms of transformation properties leaves him feeling a bit cheated. The problem here is really one of semantics. It would be clearer not to speak of "tensors in general," but instead to define "tensor character" and speak of given quantities as having or lacking a certain tensor character. All one means when one speaks of a tensor is a quantity which possesses tensor character. We conclude this section by discussing three physical quantities which have tensor character: the moment-of-interia tensor, the electric-polarization tensor, and the multipole tensor.

Example 1.6. *The moment of inertia tensor.* If a rigid body with continuous distribution of mass having density $\rho(\mathbf{r})$ rotates about one fixed point in the body, the total angular momentum about that point is

$$\mathbf{L} = \int \rho(\mathbf{r} \times \mathbf{v})\, d\tau ,$$

where $\mathbf{r}$ is the radius vector from the stationary point to the mass element $dm = \rho\, d\tau$, and $\mathbf{v}$ = the velocity of dm relative to the stationary point. It follows from kinematical considerations that $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$, where $\boldsymbol{\omega}$ is the angular velocity of the rigid body. Therefore

$$\begin{aligned} \mathbf{L} &= \int \rho[\mathbf{r} \times (\boldsymbol{\omega} \times \mathbf{r})]\, d\tau \\ &= \int \rho[\boldsymbol{\omega} r^2 - \mathbf{r}(\boldsymbol{\omega}\cdot\mathbf{r})]\, d\tau . \end{aligned}$$

Note that $\mathbf{L}$ is not parallel to $\boldsymbol{\omega}$ unless $\boldsymbol{\omega}$ is perpendicular to $\mathbf{r}$.

The inertia tensor, I_{ij}, is defined as

$$I_{ij} = \int \rho[r^2\delta_{ij} - r_i r_j]\, d\tau . \tag{1.98}$$

Note that it really is a tensor, as it is the sum of products of vectors. In fact, it is a symmetric second-rank tensor. As we shall see in chapter 4, this means that this tensor can be diagonalized—i.e., principal axes and principal moments of inertia can be found. In terms of the inertia tensor I_{ij}, we can rewrite the earlier expression for $\mathbf{L}$ as

$$\begin{aligned} I_{ij}\omega_j &= \int \rho[r^2\delta_{ij}\omega_j - r_i r_j \omega_j]\, d\tau \\ &= \int \rho[r^2\omega_i - r_i(\mathbf{r}\cdot\boldsymbol{\omega})]\, d\tau = L_i . \end{aligned}$$

Goldstein writes this equation in the symbolic form $\mathbf{L} = \mathbf{I}\cdot\boldsymbol{\omega}$, where $\mathbf{I}$ is an operator on $\boldsymbol{\omega}$. $\mathbf{I}$ is not a scalar that merely changes the magnitude of $\boldsymbol{\omega}$; it is a tensor, and thus this tensor may be viewed as changing both the magnitude and the direction of $\boldsymbol{\omega}$ in relating $\boldsymbol{\omega}$ to $\mathbf{L}$.

Example 1.7. *The electric polarization tensor.* The polarization vector $\mathbf{P}$ is related to the electric displacement vector $\mathbf{D}$ by

$$\mathbf{D} = \varepsilon_0 \mathbf{E} + \mathbf{P} .$$

If $\mathbf{P}$ is parallel to $\mathbf{E}$, we write $\mathbf{P} = \varepsilon_0 \chi \mathbf{E}$ (where χ is the electric susceptibility), so $\mathbf{D} = \varepsilon_0(1 + \chi)\,\mathbf{E} = \varepsilon_0 k \mathbf{E}$, where k is the dielectric constant. But $\mathbf{P}$ will only be parallel to $\mathbf{E}$ if the dielectric material polarizes isotropically. This can hardly be expected in crystals with molecular structures of low symmetry. In crystals of symmetry lower than cubic, the relation between each of the components of $\mathbf{P}$ and of the $\mathbf{E}$-vector is still linear, but the constants of proportionality in the various directions are not necessarily the same. The special case of $\mathbf{P}$ parallel to $\mathbf{E}$ will occur in gases, liquids, amorphous solids, Jello ® puddings, and cubic crystals. If this is not the case, the vector proportionality between $\mathbf{P}$ and $\mathbf{E}$ must be replaced by a tensor equation that allows for different constants of proportionality in different directions of $\mathbf{E}$. We have

$$P_i = \sum_{j=1}^{3} \varepsilon_0 \chi_{ij} E_j ,$$

where χ_{ij} is the susceptibility tensor.

Example 1.8. *The multipole-moment tensor.* The multipole tensor arises in the multipole expansion of the potential $\varphi(\mathbf{r})$ due to an arbitrary localized charge distribution $\rho(\mathbf{r}')$. From potential theory (e.g., Panofsky and Phillips, Chapter 1), we know that if div $\mathbf{E} = \rho$, and if curl $\mathbf{E} = 0$ (electrostatics in free space), then $\mathbf{E} = -\nabla\varphi$, where

$$\varphi(\mathbf{r}) = \int_{\tau'} \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}\, d\tau' . \tag{1.99}$$

The charge of density $\rho(\mathbf{r}')$ is localized in volume τ'. To calculate the field at points $\mathbf{r}$, remote from the source—i.e., for $|\mathbf{r}| > |\mathbf{r}'|$—we expand $1/R$ in powers of r'/r:

$$\begin{aligned}
\frac{1}{R} &\equiv |\mathbf{r} - \mathbf{r}'|^{-1} = (r^2 - 2\mathbf{r}\cdot\mathbf{r}' + r'^2)^{-1/2} \\
&= \frac{1}{r}\left[1 + \left(-\frac{2\mathbf{r}\cdot\mathbf{r}'}{r^2} + \frac{r'^2}{r^2}\right)\right]^{-1/2} \\
&= \frac{1}{r}\left[1 - \frac{1}{2}\left(-\frac{2\mathbf{r}\cdot\mathbf{r}'}{r^2} + \frac{r'^2}{r^2}\right) + \frac{1}{2}\cdot\frac{3}{4}\left(-\frac{2\mathbf{r}\cdot\mathbf{r}'}{r^2} + \frac{r'^2}{r^2}\right)^2 + \cdots\right] \\
&= \frac{1}{r}\left[1 + \frac{\mathbf{r}\cdot\mathbf{r}'}{r^2} - \frac{1}{2}\frac{r'^2}{r^2} + \frac{3}{8}\left(\frac{4(\mathbf{r}\cdot\mathbf{r}')^2}{r^4}\right) + \cdots\right] \\
&= \frac{1}{r}\left[1 + \frac{\mathbf{r}\cdot\mathbf{r}'}{r^2} + \frac{1}{2}\left(\frac{3(\mathbf{r}\cdot\mathbf{r}')^2}{r^4} - \frac{r'^2}{r^2}\right)\right. \\
&\qquad \left. + \text{ terms of order } \left(\frac{r'}{r}\right)^3 \text{ and higher.}\right]
\end{aligned}$$

Table 1.2.

Symbol	Definition	Name	Tensor character
q	$\int_{\tau'} \rho(r')\, d\tau'$	monopole (charge)	scalar
$\mathbf{p}$	$\int_{\tau'} \mathbf{r}'\rho(\mathbf{r}')\, d\tau'$	dipole	vector
Q_{ij}	$\int_{\tau'} (3x_i'x_j' - \delta_{ij}r'^2)\rho(\mathbf{r}')\, d\tau'$	quadrupole	second-rank symmetric tensor

Note at this point that if we put $\mathbf{r}\cdot\mathbf{r}' = rr'\cos\theta$, the multipole expansion can be written as

$$\frac{1}{R} = \frac{1}{r}\left[1 + \left(\frac{r'}{r}\right)\cos\theta + \left(\frac{r'}{r}\right)^2 \frac{3\cos^2\theta - 1}{2} + \cdots\right]. \tag{1.100}$$

In Chapter 6 we shall prove that, for $r' < r$,

$$\frac{1}{R} \equiv \frac{1}{|\mathbf{r} - \mathbf{r}'|} = \frac{1}{r}\sum_{l=0}^{\infty}\left(\frac{r'}{r}\right)^l P_l(\cos\theta)\ , \tag{1.101}$$

where the $P_l(\cos\theta)$ are the Legendre polynominals:

$$\begin{aligned} P_0(\cos\theta) &= 1\ , \\ P_1(\cos\theta) &= \cos\theta\ , \\ P_2(\cos\theta) &= \tfrac{1}{2}(3\cos^2\theta - 1)\ , \\ P_3(\cos\theta) &= \tfrac{1}{2}(5\cos^3\theta - 3\cos\theta)\ , \text{ etc.} \end{aligned} \tag{1.102}$$

We have proved by direct expansion that the first three terms of Eq. (1.101) are correct. If we now substitute the expansion of R^{-1} into the expression for $\varphi(r)$, we have

$$\begin{aligned} \varphi(r) = {} & \frac{1}{r}\int_{\tau'} \rho(\mathbf{r}')\, d\tau' + \frac{\mathbf{r}}{r^3}\cdot\int_{\tau'} \mathbf{r}'\rho(\mathbf{r}')\, d\tau' \\ & + \frac{1}{2}\frac{x_i x_j}{r^5}\int_{\tau'} [3x_i'x_j' - \delta_{ij}r'^2]\rho(\mathbf{r}')\, d\tau' + \cdots . \end{aligned} \tag{1.103}$$

The first three multipoles are defined in Table 1.2.

Higher-order poles are defined by expanding the potential to higher order. It is clear from the definition of the quadrupole-moment tensor that it is symmetric. The *trace* of a tensor T_{ij} is defined as tr $(T_{ij}) \equiv \sum_i T_{ii}$ = the sum of the diagonal elements. Note that

$$\begin{aligned} \operatorname{tr} Q_{ij} &= \sum_i Q_{ii} = \sum_i \int_{\tau'} (3x_i'x_i' - \delta_{ii}r'^2)\, \rho(\mathbf{r}')\, d\tau' \\ &= \int_{\tau'} (3r'^2 - 3r'^2)\, \rho\, d\tau' = 0\ . \end{aligned}$$

So the trace of the quadrupole-moment tensor is identically zero. Knowledge of the quadrupole moment is useful in determining the shape of nuclei and thus obtaining information on nuclear forces. In the multipole expansion, the first term is just the total nuclear charge and the second term is the dipole moment, which can be shown to vanish for quantum-mechanical systems in stationary states. The third term is a quadrupole moment. By measuring it and then fitting the result to a model which predicts such a quadrupole moment, we learn the shape of the nucleus in question. I. I. Rabi and co-workers showed that the deuteron possesses a nonzero quadrupole moment such that it must have the shape of a prolate ellipsoid, 50% longer than it is wide. This implies a *non-central nuclear force* between the neutron and the proton, a fact of great importance.

PROBLEMS

1. Prove by vector methods that the diagonals of a rhombus are orthogonal.
2. A particle moves in the $x_1 x_2$-plane so that its position vector is given by
$$\mathbf{r} = a \cos \omega t \mathbf{e}_1 + a \sin \omega t \mathbf{e}_2 \qquad (\omega = \text{const}) .$$
Show that
   a) the velocity $\mathbf{v}$ is perpendicular to $\mathbf{r}$;
   b) the acceleration $\mathbf{a}$ is directed toward the origin; find its magnitude;
   c) $\mathbf{L} = \mathbf{r} \times m\mathbf{v}$ is a constant vector; find its magnitude.
3. Prove from simple vector arguments and the conservation laws of mechanics that in an elastic collision between two particles of equal mass, one initially at rest, the velocity vectors after collision are perpendicular (if nonzero).
4. A charged particle moves in a region of space where there is a uniform magnetic field $\mathbf{B}$ (in the z-direction say,) and a uniform electric field $\mathbf{E}$ in the yz-plane. Write the equations of motion for the particle. Then solve them, taking for initial conditions (at $t = 0$) $x = y = z = 0$; $\dot{x} = \dot{x}_0$, $\dot{y} = \dot{z} = 0$. Show that there is a drift velocity equal to E_y/B in the positive x-direction. If $E_y = 0$, then the orbit becomes a spiral about a line parallel to the z-axis.
5. Prove that $\varepsilon_{ijk}\varepsilon_{ijk} = 6$.
6. Prove the vector identities:
   a) $\text{grad}\,(\mathbf{A}\cdot\mathbf{B}) = (\mathbf{A}\cdot \text{grad})\,\mathbf{B} + (\mathbf{B}\cdot \text{grad})\,\mathbf{A} + \mathbf{A}\,\text{curl}\,\mathbf{B} + \mathbf{B}\,\text{curl}\,\mathbf{A}$;
   b) $\nabla\cdot(\mathbf{U} \times \mathbf{V}) = \mathbf{V}\cdot(\nabla \times \mathbf{U}) - \mathbf{U}\cdot(\nabla \times \mathbf{V})$.
7. Prove Green's theorem, that is,
$$\int_\tau (\varphi\nabla^2\psi - \psi\nabla^2\varphi)\, d\tau = \int_\sigma (\varphi\nabla\psi - \psi\nabla\varphi)\cdot d\boldsymbol{\sigma} .$$
Hint: Apply Gauss's divergence theorem to the vectors $\mathbf{A} = \varphi\nabla\psi$ and $\mathbf{B} = \psi\nabla\varphi$.
8. If the electric potential of a point charge were
$$\varphi(r) = \frac{q}{r^{1-\varepsilon}}, \qquad \varepsilon \ll 1 \qquad \text{(positive or negative)}$$
instead of q/r, many of the results of physics would be altered. Calculate for $r \neq 0$:
   a) $\mathbf{E} = -\nabla\varphi$, b) $\nabla\cdot\mathbf{E} = -\nabla^2\varphi$, c) $\nabla \times \mathbf{E}$,
   d) Derive the analog of Gauss's law for a spherical region of radius R.

Examine the limits as $\varepsilon \to 0$, and compare with the usual Coulomb potential. One of the consequences of such a potential is discussed in Problem 8, Panofsky and Phillips, Chapter 1.

9. A physical system known as a vector meson is described by the three vector fields **E**, **H**, **A** and a real scalar field V, which satisfy the following equations:

$$\nabla \cdot \mathbf{E} = -\mu^2 V$$

$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{E}}{\partial t} - \mu^2 \mathbf{A} \qquad (\mu^2 \text{ is a positive constant})$$

$$\mathbf{H} = \nabla \times \mathbf{A}\,, \qquad \mathbf{E} = \frac{\partial \mathbf{A}}{\partial t} - \nabla V$$

a) Show that $\nabla \cdot \mathbf{A} + \dfrac{\partial V}{\partial t} = 0$

b) Show that V satisfies the equation:

$$\nabla^2 V - \mu^2 V - \frac{\partial^2 V}{\partial t^2} = 0\,.$$

c) When V is time-independent, it satisfies $\nabla^2 V - \mu^2 V = 0$. If this is the case inside a closed surface and if V vanishes on the closed surface, show that $V = 0$ inside.

10. a) What class of physical phenomena are described by Helmholtz's equation,

$$\nabla^2 \varphi + k^2 \varphi = 0\,?$$

b) Interpret this equation physically in the light of the discussion of the physical meaning of the Laplacian, and square your interpretation with the answer to part (a).

11. a) Derive the law of conservation of charge from Maxwell's equations.

b) Express the electric field in terms of the scalar potential φ and the vector potential **A** where $\mathbf{B} = \operatorname{curl} \mathbf{A}$. (*Note*: For the general time-dependent case, **E** is not equal to grad φ alone.)

12. A second-rank antisymmetric tensor F_{ij} is characterized by three independent quantities. Consequently, a vector H_k can be associated with such a tensor. Prove that $H_k = \frac{1}{2}\varepsilon_{ijk} F_{ij}$ *if and only if* $F_{ij} = \varepsilon_{ijk} H_k$.

13. *The electromagnetic field tensor in free space.* Problems 11 and 12 have laid the groundwork for the four-dimensional theory of the electromagnetic field that will unfold as this problem is done. We shall use Gaussian units. To begin, if we identify the vector H_k of Problem 11 with the magnetic field, then problem 12 relates H_k to a second-rank antisymmetric tensor, F_{ij}.

a) Prove

$$F_{ij} = \frac{\partial A_j}{\partial x_i} - \frac{\partial A_i}{\partial x_j} \equiv \partial_i A_j - \partial_j A_i$$

where $\operatorname{curl} \mathbf{A} = \mathbf{H}$.

b) Show directly from (a) that F_{ij} is antisymmetric.

So far, F_{ij} is three-dimensional and has $3^2 = 9$ components. We now extend it to four dimensions.

c) Make the identifications $x_4 = ict$ and $A_4 = i\varphi$, where $i = \sqrt{-1}$ and φ is the electrostatic potential. Prove that $F_{4j} = -F_{j4} = iE_j$ $(j = 1, 2, 3)$, using F_{ij} as expressed in (a) and problem 11(b).

d) Verify that $F_{ii} = 0$ $(i = 1, 2, 3, 4)$.
Thus F_{ij} is a four-dimensional antisymmetric second-rank tensor with $4^2 = 16$ components. The four diagonal elements are zero. Only six of the remaining 12 elements are independent, expressing the fact that **E** and **H** have a total of six components.

e) Write out F_{ij} as a 4 by 4 array explicitly in terms of the components of **E** and **H**. At this point you might want to see how you are doing. You should obtain

$$F_{ij} = \begin{bmatrix} 0 & H_z & -H_y & -iE_x \\ -H_z & 0 & H_x & -iE_y \\ H_y & -H_x & 0 & -iE_z \\ iE_x & iE_y & iE_z & 0 \end{bmatrix}.$$

f) All four of Maxwell's equations can be expressed in two equations written in terms of the electromagnetic field tensor:

i. $\partial_l F_{ik} + \partial_i F_{kl} + \partial_k F_{li} = 0$,
ii. $\partial_k F_{ik} = (4\pi/c)\, J_i$,

where $J_i = (\mathbf{J})_i$ for $i = 1, 2, 3$ and $J_4 = ic\rho$ (ρ = electric-charge density). In these two equations i, k, and l extend from 1 to 4. Prove that these equations subsume Maxwell's equations, and give nothing more. (This will involve considering different combinations of subscripts, etc.)

g) Derive the four-dimensional form of the equation of conservation of charge directly from Maxwell's equation (ii) in part (f). Interpret it in four-dimensional form.

h) Show that the Lorentz force law is

$$f_i = \frac{1}{c} F_{ik} J_k\,, \qquad i = 1, 2, 3\,,$$

where f_i are the three components of the Lorentz force. Interpret the quantity icf_4.

14. Prove that the trace of the electromagnetic energy-momentum tensor, T_{ik}, is zero, where

$$T_{ik} = \frac{1}{4\pi}\left(F_{il}F_{kl} - \frac{1}{4}F_{lm}^2\delta_{ik}\right);$$

F_{ij} is the electromagnetic field tensor of Problem 13.

15. Prove that the dipole moment of a charge distribution is independent of the origin of coordinates if the total charge is zero.

16. a) A charge q is uniformly distributed upon a circle that has radius R, is concentric with the z-axis, and lies in the plane $z = 0$. A charge $(-q)$ is located at the center of the circle. Find (i) the electric monopole moment (ii) the electric-dipole moment vector and (iii) the electric-quadrupole moment tensor, expressing the latter in an appropriate 3 by 3 array. Using the symmetry of the quadrupole tensor and the fact that its trace vanishes, as well as the inherent symmetry of the physical system under consideration, one can reduce the labor enormously.

b) Determine for this charge distribution the multipole expansion for the potential at distances r from the origin for which $(r/R) \gg 1$; carry the expansion through the quadrupole contribution, and express the spatial dependence in the final result in terms of the spherical coordinates r, θ, φ.